

7 Series FPGAs XADC Dual 12-Bit 1MSPS Analog-to-Digital Converter

User Guide

UG480 (v1.1) March 28, 2011



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Revision History

The following table shows the revision history for this document.

Date	Version	Revision
03/01/11	1.0	Initial Xilinx release.
03/28/11	1.1	Added "Dual 12-Bit MSPS Analog-to-Digital Converter" to document title. Modified first paragraph and added second paragraph in Chapter 1, Introduction and Quick Start . Added Table 1-1 . Removed "Thermal Diode (DXP and DXN)" section from Chapter 4.

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About This Guide

Xilinx® 7 series FPGAs include three unified FPGA families that are all designed for lowest power to enable a common design to scale across families for optimal power, performance, and cost. The Artix™-7 family is optimized for lowest cost and absolute power for the highest volume applications. The Virtex®-7 family is optimized for highest system performance and capacity. The Kintex™-7 family is an innovative class of FPGAs optimized for the best price-performance. This guide serves as a technical reference describing the 7 series FPGAs XADC, a dual 12-bit, 1 MSPS analog-to-digital converter with on-chip sensors.

This *7 Series FPGAs XADC User Guide*, part of an overall set of documentation on the 7 series FPGAs, is available on the Xilinx website at www.xilinx.com/7.

Guide Contents

This manual contains these chapters:

- [Chapter 1, Introduction and Quick Start](#)
- [Chapter 2, Analog-to-Digital Converter](#)
- [Chapter 3, XADC Register Interface](#)
- [Chapter 4, XADC Operating Modes](#)
- [Chapter 5, XADC Timing](#)
- [Chapter 6, Application Guidelines](#)

Additional Resources

To find additional documentation, see the Xilinx website at:

<http://www.xilinx.com/support/documentation/index.htm>.

To search the Answer Database of silicon, software, and IP questions and answers, or to create a technical support WebCase, see the Xilinx website at:

<http://www.xilinx.com/support>.

References

Additional material useful to this document can be found here:

- UG772, *LogiCORE™ IP XADC Wizard User Guide*

Introduction and Quick Start

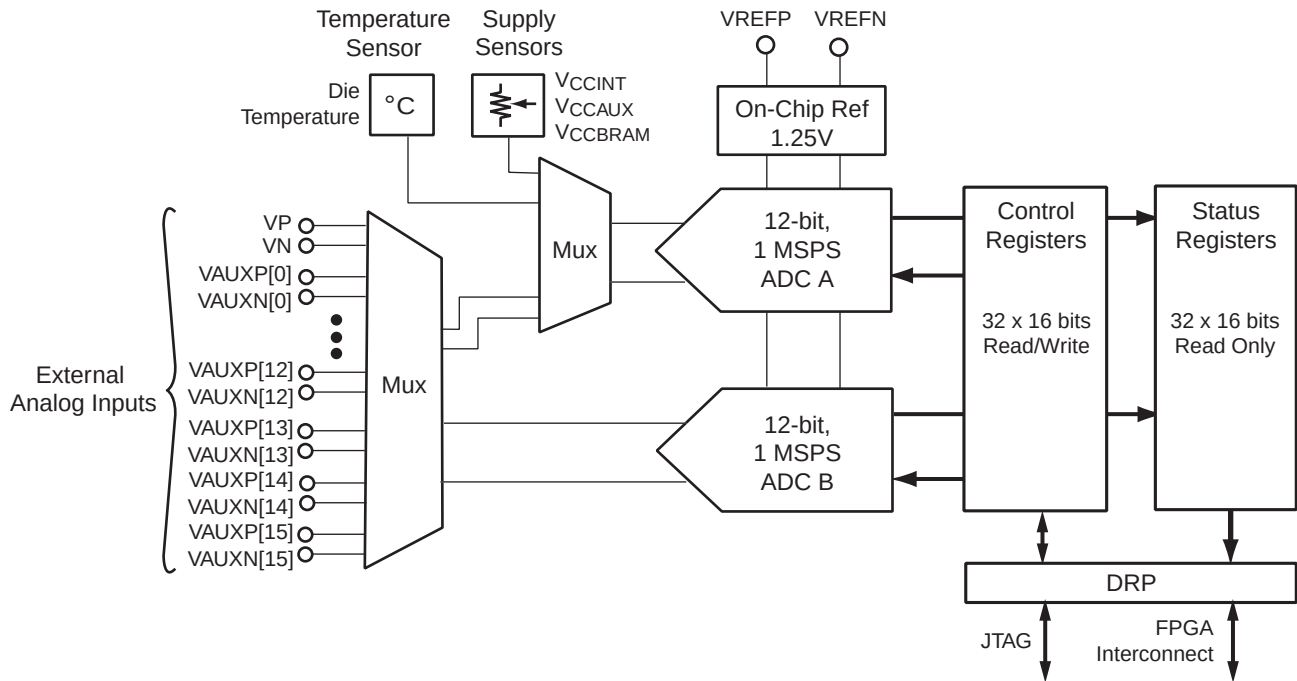
This chapter provides a brief overview of the Xilinx® 7 series FPGAs XADC functionality. The XADC is available in all Artix™, Kintex™, and Virtex® family members.

The XADC is the basic building block that enables agile mixed signal functionality which is new to 7 series FPGAs. By combining high quality analog blocks with the flexibility of programmable logic, it is possible to craft customized analog interfaces for a wide range of applications. Please see <http://www.xilinx.com/ams> for more information.

This chapter contains only key information to allow a basic understanding of the XADC block. With this introduction, users can learn the pinout requirements and determine how to instantiate basic functionality in their designs. Subsequent chapters provide more detailed descriptions of the XADC functionality.

XADC Overview

The XADC includes a dual 12-bit, 1 Mega sample per second (MSPS) ADC and on-chip sensors. These ADCs are fully tested and specified (see the respective 7 series FPGAs data sheet). The ADCs provide a general-purpose, high-precision analog interface for a range of applications. [Figure 1-1](#) shows a block diagram of the XADC. The dual ADCs support a range of operating modes, for example, externally triggered and simultaneous sampling on both ADCs (see [Chapter 4, XADC Operating Modes](#)) and various analog input signal types, for example, unipolar and differential (see [Chapter 2, Analog-to-Digital Converter](#)). The ADCs can access up to 17 external analog input channels.



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Figure 1-1: XADC Block Diagram

The XADC also includes a number of on-chip sensors that support measurement of the on-chip power supply voltages and die temperature. The ADC conversion data is stored in dedicated registers called status registers. These registers are accessible via the FPGA interface using a 16-bit synchronous read and write port called the Dynamic Reconfiguration Port (DRP). ADC conversion data is also accessible via the JTAG TAP. In the latter case, users are not required to instantiate the XADC because it is a dedicated interface that uses the existing FPGA JTAG infrastructure. As discussed later, if the XADC is not instantiated in a design, the device operates in a predefined mode (called *default mode*) that monitors on-chip temperature and supply voltages.

XADC operation is user defined by writing to the control registers using either the DRP or JTAG interface. It is also possible to initialize these register contents when the XADC is instantiated in a design using the block attributes.

System Monitor Support

For Virtex-5 and Virtex-6 FPGA System Monitor users, the XADC functionality is fully backward compatible with legacy System Monitor designs. The XADC functionality and interface will be familiar to those who have previously designed with the System Monitor. System Monitor designs are automatically retargeted to the XADC site by the software tools.

XADC Pinout Requirements

Dedicated Package Pins

Figure 1-2 shows the basic pinout requirements for the XADC. There are two recommended configurations. On the left, the XADC is powered from V_{CCAUX} (1.8V) and uses an external 1.25V reference source. The external reference delivers the best performance in terms of accuracy and thermal drift. A ferrite bead is used to isolate the ground reference for the analog circuits and system ground (see Chapter 6, Application Guidelines for more information). Shared or common ground impedance is the most common way to introduce unwanted noise into analog circuits.

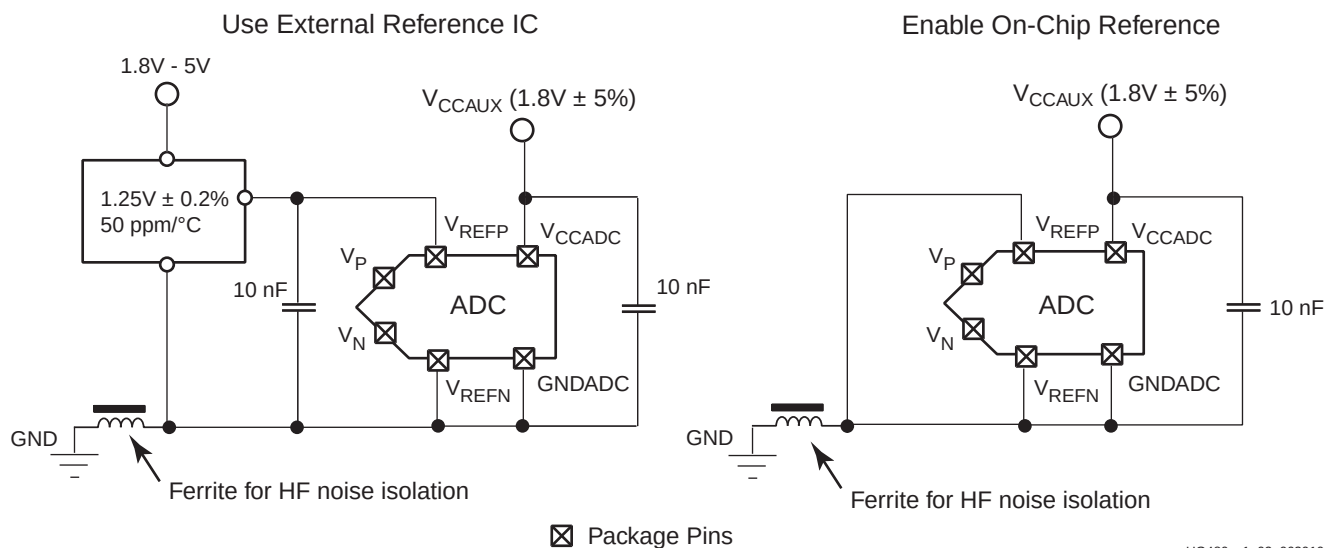


Figure 1-2: XADC Pinout Requirements

To enable the on-chip reference source, the V_{REFP} pin must be connected to ground as shown on the right of Figure 1-2. Where only basic on-chip thermal and supply monitoring is required, using the on-chip reference provides good performance. Users should consult the respective 7 series FPGAs data sheet to see the accuracy specifications when using the external and on-chip reference sources. Table 1-1 lists the pins associated with the XADC and the recommended connectivity.

Table 1-1: XADC Package Pins

Package Pin	Type	Description
V _{CCADC_0}	Power supply	This is the analog supply pin for the ADCs and other analog circuits in the XADC. It can be tied to the 1.8V V _{CCAUX} supply however in a mixed-signal system the supply should be connected to a separate 1.8V analog if available. See Analog Power Supply and Ground (V_{CCADC} and GNDADC) , page 65 for more information. This pin should never be tied to GND. The pin should be tied to V _{CCAUX} even if the XADC is not being used.
GNDADC_0	Power supply	This is the ground reference pin for the ADCs and other analog circuits in the XADC. It can be tied to the system ground via an isolating ferrite bead as shown in Figure 1-2 . In a mixed-signal system this pin should be tied to an analog ground plane if available, in which case the ferrite bead is not required. See Analog Power Supply and Ground (V_{CCADC} and GNDADC) , page 65 for more information. This pin should always be tied to GND even if the XADC is not being used.
V _{REFP_0}	Reference voltage	This pin should be tied to an external 1.25V accurate reference source ($\pm 0.2\%$) for best performance of the ADCs. It should be treated as an analog signal that together with the V _{REFN_0} signal provides a differential 1.25V voltage. By connecting this pin to GND (see Figure 1-2) an on-chip reference source ($\pm 1\%$) is activated. This pin should always be connected to GND if an external reference is not supplied. See Reference Inputs (V_{REFP} and V_{REFN}) , page 65 for more information.
V _{REFN_0}	Reference voltage	This pin should be tied to GND pin of an external 1.25V accurate reference source ($\pm 0.2\%$) for best performance of the ADCs. It should be treated as an analog signal that together with the V _{REFP_0} signal provides a differential 1.25V voltage. This pin should always be connected to GND even if an external reference is not supplied. See Reference Inputs (V_{REFP} and V_{REFN}) , page 65 for more information.
V _{P_0}	Analog input	This is the positive input terminal of the dedicated differential analog input channel (VP/VN). The analog input channels are very flexible and support multiple analog input signal types. For more information see Analog Inputs , page 24. This pin should be connected to GND if not used.
V _{N_0}	Analog input	This is the negative input terminal of the dedicated differential analog input channel (VP/VN). The analog input channels are very flexible and support multiple analog input signal types. For more information see Analog Inputs , page 24. This pin should be connected to GND if not used.
ADP0 to ADP15 ⁽¹⁾	Analog inputs/ digital I/O	These are multi-function pins that can support analog inputs or can be used as regular digital IO (see Figure 1-1). These pins support up to 16 positive input terminals of the differential auxiliary analog input channels (V _{AUXP} /V _{AUXN}). The analog input channels are very flexible and support multiple analog input signal types. For more information see Analog Inputs , page 24. When not being used as analog input these pins can be treated like any other digital IO.

Table 1-1: XADC Package Pins

Package Pin	Type	Description
ADN0 to ADN15 ⁽¹⁾	Analog inputs/ digital I/O	These are multi-function pins that can support analog inputs or can be used as regular digital IO (see Figure 1-1). These pins support up to 16 negative input terminals of the differential auxiliary analog input channels (V_{AUXP}/V_{AUXN}). The analog input channels are very flexible and support multiple analog input signal types. For more information see Analog Inputs, page 24 . When not being used as analog input these pins can be treated like any other digital IO.

Notes:

1. Auxiliary Channels 6, 7, 13, 14, and 15 are not supported in Kintex-7 devices. The auxiliary analog input channels are located in bank 15 and 35.

[Chapter 6, Application Guidelines](#), should be consulted before commencing any PC board layout. Board layout and external component choices can greatly impact the performance of the ADCs.

External Analog Inputs

Apart from a single dedicated analog input pair (V_P/V_N), the external analog inputs use dual-purpose I/O. These FPGA digital I/Os are individually nominated as analog inputs when the XADC is instantiated in a design. This document refers to these analog inputs as auxiliary analog inputs. A maximum of 16 auxiliary analog inputs are available. The auxiliary analog inputs are enabled by connecting the analog inputs on the XADC primitive to the top level of the design. When allocated as analog inputs, these package balls are unavailable as digital I/Os.

All analog input channels are differential and require two package balls. Typically, the auxiliary analog inputs are allocated evenly over banks 15 and 35. The pinout information for the 7 series FPGA family indicates which I/Os and banks for a particular device and package combination are analog capable by appending an $ADxP$ or $ADxN$ suffix to the I/O name. For example, auxiliary analog input channel 8 has associated package ball names ending with $AD8P$ and $AD8N$. See [UG475, 7 Series FPGAs Packaging and Pinout Specification](#) for more information. The auxiliary analog inputs have a fixed package ball assignment and cannot be moved.

Note: Auxiliary channels 6, 7, 13, 14, and 15 are not supported on Kintex-7 devices.

Instantiating the XADC

As mentioned previously, it is not necessary to instantiate the XADC in a design to access the on-chip monitoring capability. However, if the XADC is not instantiated in a design, the only way to access this information is via the JTAG TAP. To allow access to the status registers (measurement results) from the FPGA logic, the XADC must be instantiated. These subsections give a brief overview of the XADC primitive (ports and attributes).

XADC Ports

Figure 1-3 shows the ports on the XADC primitive, and Table 1-2 describes the functionality of the ports.

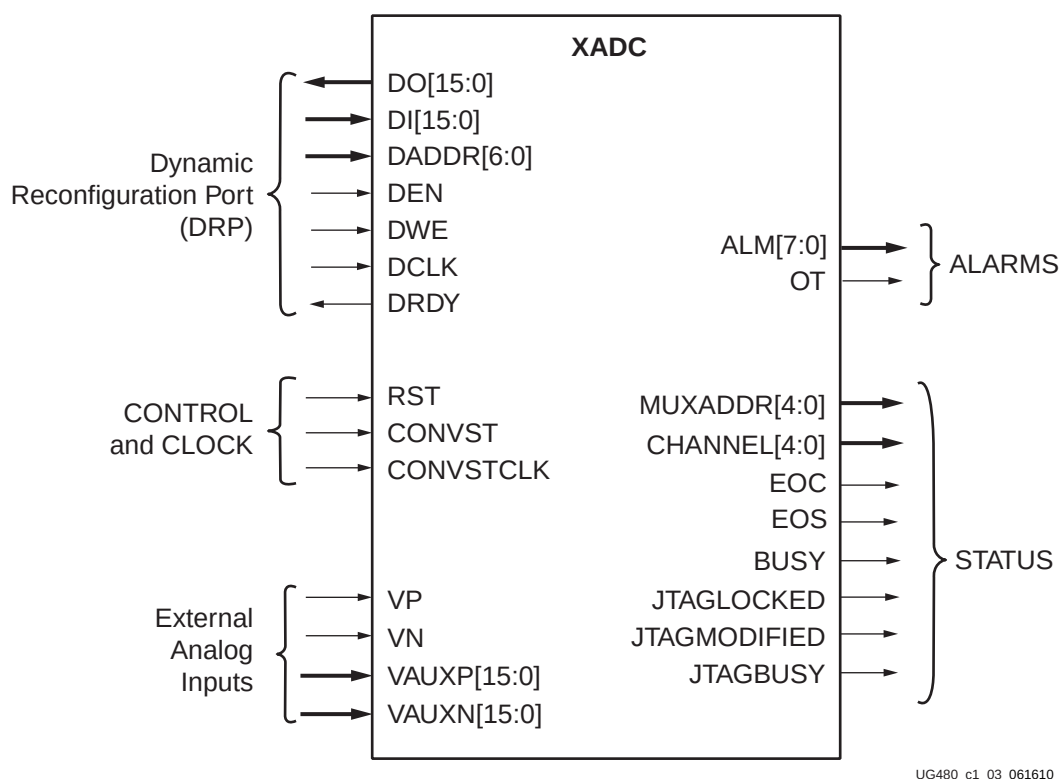


Figure 1-3: XADC Primitive Ports

Table 1-2: XADC Port Descriptions

Port	I/O	Description
DI[15:0]	Inputs	Input data bus for the DRP. ⁽¹⁾
DO[15:0]	Outputs	Output data bus for the DRP. ⁽¹⁾
DADDR[6:0]	Input	Address bus for the DRP. ⁽¹⁾
DEN ⁽²⁾	Input	Enable signal for the DRP. ⁽¹⁾
DWE ⁽²⁾	Input	Write enable for the DRP. ⁽¹⁾
DCLK	Input	Clock input for the DRP. ⁽¹⁾
DRDY ⁽²⁾	Output	Data ready signal for the DRP. ⁽¹⁾
RESET ⁽²⁾	Input	Reset signal for the XADC control logic.
CONVST ⁽³⁾	Input	Convert start input. This input controls the sampling instant on the ADC(s) input and is only used in event mode timing (see Event-Driven Sampling , page 61). This input comes from the general-purpose interconnect in the FPGA logic.

Table 1-2: XADC Port Descriptions (Cont'd)

Port	I/O	Description
CONVSTCLK ⁽³⁾	Input	Convert start clock input. This input is connected to a clock net. Like CONVST, this input controls the sampling instant on the ADC(s) inputs and is only used in event mode timing. This input comes from the local clock distribution network in the FPGA logic. Thus, for the best control over the sampling instant (delay and jitter), a global clock input can be used as the CONVST source.
V _P , V _N	Input	One dedicated analog input pair. The XADC has one pair of dedicated analog input pins that provide a differential analog input. When designing with the XADC feature but not using the dedicated external channel of V _P and V _N , the user should connect both V _P and V _N to analog ground.
VAUXP[15:0], VAUXN[15:0]	Inputs	Sixteen auxiliary analog input pairs. In addition to the dedicated differential analog input, the XADC can access 16 differential analog inputs by configuring digital I/O as analog inputs. These inputs can also be enabled preconfiguration via the JTAG port (see DRP JTAG Interface, page 39).
ALM[0] ⁽²⁾	Output	XADC temperature sensor alarm output.
ALM[1] ⁽²⁾	Output	XADC V _{CCINT} sensor alarm output.
ALM[2] ⁽²⁾	Output	XADC V _{CCAUX} sensor alarm output.
ALM[3] ⁽²⁾	Output	XADC V _{CCBRAM} sensor alarm output.
ALM[6:4]	Outputs	Not defined.
ALM[7] ⁽²⁾	Output	Logic OR of bus ALM[6:0]. Can be used to flag the occurrence of any alarm.
OT ⁽²⁾	Output	Over-Temperature alarm output.
MUXADDR[4:0]	Outputs	These outputs are used in external multiplexer mode. They indicate the address of the next channel in a sequence to be converted. They provide the channel address for an external multiplexer (see External Multiplexer Mode, page 52).
CHANNEL[4:0]	Outputs	Channel selection outputs. The ADC input MUX channel selection for the current ADC conversion is placed on these outputs at the end of an ADC conversion.
EOC ⁽²⁾	Output	End of Conversion signal. This signal transitions to active High at the end of an ADC conversion when the measurement is written to the status registers (see Chapter 5, XADC Timing).
EOS ⁽²⁾	Output	End of Sequence. This signal transitions to active High when the measurement data from the last channel in an automatic channel sequence is written to the status registers (see Chapter 5, XADC Timing).

Table 1-2: XADC Port Descriptions (Cont'd)

Port	I/O	Description
BUSY ⁽²⁾	Output	ADC busy signal. This signal transitions High during an ADC conversion. This signal also transitions High for an extended period during an ADC or sensor calibration.
JTAGLOCKED ⁽²⁾	Output	Indicates that a DRP port lock request has been made by the JTAG interface (see DRP JTAG Interface, page 39). This signal is also used to indicate that the DRP is ready for access (when Low).
JTAGMODIFIED ⁽²⁾	Output	Used to indicate that a JTAG write to the DRP has occurred.
JTAGBUSY ⁽²⁾	Output	Used to indicate that a JTAG DRP transaction is in progress.

Notes:

1. The DRP is the interface between the XADC and FPGA. All XADC registers can be accessed from the FPGA logic using this interface. For more details on the timing for these DRP signals, consult the *7 Series FPGAs Configuration User Guide*.
2. Active-High signal.
3. Rising edge triggered signal.

XADC Attributes

The block diagram in [Figure 1-1](#) shows the control registers that define the operation of the XADC. The control registers are a set of 32 16-bit registers. As mentioned, these registers can be read and written via the DRP or JTAG ports. It is also possible to initialize the contents of these registers during the configuration of the FPGA. This enables the XADC to start operating in a user-defined mode after configuration is complete. There are 32 attributes associated with the XADC primitive that allow users to initialize these registers. [Table 1-3](#) lists these attributes. The attributes are called INIT_xx, where xx corresponds to the hexadecimal address of the register on the DRP. For example, INIT_40 corresponds to the first control register at address 40h on the DRP.

Table 1-3: XADC Primitive Attributes

Attribute	Name	Control Register Address	Description
INIT_40	Configuration Register 0	40h	XADC configuration registers (see Control Registers, page 34).
INIT_41	Configuration Register 1	41h	
INIT_42	Configuration Register 2	42h	
INIT_43 to INIT_47	Test Registers	43h to 47h	XADC Test registers for factory use only. The default initialization is 0000h.

Table 1-3: XADC Primitive Attributes (Cont'd)

Attribute	Name	Control Register Address	Description
INIT_48 to INIT_4F	Sequence Registers	48h to 4Fh	Sequence registers used to program the XADC Channel Sequencer function (see Automatic Channel Sequencer , page 45).
INIT_50 to INIT_5F	Alarm Limit Registers	50h to 5Fh	Alarm threshold registers for the XADC alarm function (see Automatic Alarms , page 55).

The XADC primitive has an attribute called SIM_MONITOR_FILE that points to the analog stimulus file. This attribute is required to support simulation. This attribute points to the path and file name of a text file that contains analog information (e.g., temperature and voltage). UNISIM and SIMPRIM use this text file during simulation. This is the only way analog signals can be introduced into a simulation of the XADC. For more information see [XADC Software Support in Chapter 6](#).

Example Instantiation

Instantiating the XADC involves connecting the required I/O (including analog inputs) to the design and optionally initializing the control registers to define the XADC operation after configuration. Alternatively, users can write to the control registers via the DRP after device configuration.

This subsection provides an example of an XADC instantiation using Verilog. First, the control registers are initialized, and then the required XADC I/Os are connected to the design. The software correctly ties off unconnected I/Os on the primitive.

This design assumes an external 50 MHz clock is used for DCLK, and the XADC is configured to monitor temperature, supply voltages, and activate alarms if safe limits are exceeded. This example is explained in detail in the [XADC Software Support](#) section.

```

module My_XADC

XADC #(
    // Initializing the XADC Control Registers
    .INIT_40(16'h0000), // Config reg 0
    .INIT_41(16'h2ef0), // Continuous Sequencer Mode, Disable unused ALMs,
    // Enable calibration
    .INIT_42(16'h0400), // Set DCLK divider to 4, ADC = 500 Ksps,
    // DCLK = 50 MHz
    .INIT_48(16'h4701), // Sequencer channel - enable on-chip sensors and
    // calibration
    .INIT_49(16'h000f), // Sequencer channel - enable aux channels 0 - 3
    .INIT_4A(16'h0000), // Sequencer Average selection
    .INIT_4B(16'h0000), // Sequencer Average selection
    .INIT_4C(16'h0000), // Sequencer Bipolar selection
    .INIT_4D(16'h0000), // Sequencer Bipolar selection
    .INIT_4E(16'h0000), // Sequencer Acq time selection
    .INIT_4F(16'h0000), // Sequencer Acq time selection
    .INIT_50(16'hb5ed), // Temp upper alarm trigger 85°C
    .INIT_51(16'h5999), // Vccint upper alarm limit 1.05V
    .INIT_52(16'hA147), // Vccaux upper alarm limit 1.89V
    .INIT_53(16'hdddd), // OT upper alarm limit 125°C
    .INIT_54(16'ha93a), // Temp lower alarm reset 60°C
    .INIT_55(16'h5111), // Vccint lower alarm limit 0.95V
    .INIT_56(16'h91Eb), // Vccaux lower alarm limit 1.71V

```

```

        .INIT_57(16'hae4e),      // OT lower alarm reset 70°C
        .INIT_58(16'h5999),      // VBRAM upper alarm limit 1.05V
        .INIT_5C(16'h5111),      // VBRAM lower alarm limit 0.95V
        .SIM_MONITOR_FILE("sensor_input.txt") // Analog Stimulus file
    )

    XADC_INST (                  // See UG for port descriptions
        .CONVST(GND_BIT),
        .CONVSTCLK(GND_BIT),
        .DADDR(DADDR_IN[6:0]),
        .DCLK(DCLK_IN),
        .DEN(DEN_IN),
        .DI(DI_IN[15:0]),
        .DWE(DWE_IN),
        .RESET(RESET_IN),
        .VAUXN(aux_channel_n[15:0]),
        .VAUXP(aux_channel_p[15:0]),
        .ALM(alm_int),
        .BUSY(BUSY_OUT),
        .CHANNEL(CHANNEL_OUT[4:0]),
        .DO(DO_OUT[15:0]),
        .DRDY(DRDY_OUT),
        .EOC(EOC_OUT),
        .EOS(EOS_OUT),
        .JTAGBUSY(),
        .JTAGLOCKED(),
        .JTAGMODIFIED(),
        .OT(OT_OUT),
        .MUXADDR(),
        .VP(VP_IN),
        .VN(VN_IN)
    );

endmodule

```

ADC and Sensors

More comprehensive information regarding the operation of the ADCs and on-chip sensors can be found in [Chapter 2, Analog-to-Digital Converter](#). This section provides a brief overview to help users to quickly interpret data read from the status registers and verify the operation of the XADC.

Analog-to-Digital Converter

The ADCs have a nominal analog input range from 0V to 1V. In unipolar mode (default), the analog inputs of the ADCs produce a full scale code of FFFh (12 bits) when the input is 1V. In bipolar mode, the ADCs use 2's complement coding and produce a full scale code of 7FFh with +0.5V input and 800h with -0.5V input. Thus, an analog input signal of 200 mV in unipolar mode produces and outputs code of 819 or 333h.

Temperature Sensor

The temperature sensor has a transfer function given by [Equation 1-1](#).

$$Temp (^{\circ}C) = \frac{ADC\ Code \times 503.975}{4096} - 273.15 \quad \text{Equation 1-1}$$

For example, ADC Code 2423 (977h) = 25°C.

The temperature sensor result can be found in Status Register 00h.

Power Supply Sensors

The XADC power supply sensors have a transfer function that generates a full scale ADC output code of FFFh with a 3V input voltage. This voltage is outside the allowed supply range, but the FPGA supply measurements map into this measurement range. Thus, $V_{CCINT} = 1V$ generates an output code of $1/3 \times 4096 = 1365 = 555h$. The XADC monitors V_{CCINT} , V_{CCAUX} , and V_{CCBRAM} . The measurement results are stored in Status Registers 01h, 02h, and 06h, respectively.

Analog-to-Digital Converter

The XADC block contains two 12-bit, 1 MSPS ADCs. These ADCs are available for use with both external analog inputs and on-chip sensors. A number of predefined operating modes are available that cover the most typical use cases for these ADCs. The various operating modes are covered in [Chapter 4, XADC Operating Modes](#). This chapter focuses on the detailed operation of the ADC and the on-chip sensors. The various configurations for the external analog inputs are also covered. All operating modes of the ADC, sensors, and analog inputs are configured using the XADC control registers. A detailed description of the control registers is covered in [Chapter 3, XADC Register Interface](#).

ADC Transfer Functions

The ADCs have transfer functions as shown in [Figure 2-1](#) and [Figure 2-2](#). These transfer functions reflect unipolar and bipolar operating modes, respectively. All on-chip sensors use the unipolar mode of operation for the ADC. Users can optionally configure the external analog input channels to operate in unipolar or bipolar modes (see [Analog Inputs](#), page 24).

For the ADCs to function as specified, the power supplies and reference options must be configured correctly. The required package ball connections are shown in [Figure 1-2](#). Other important aspects to ensure optimal ADC performance are the PCB layout and external component selection. These issues are covered in [Chapter 6, Application Guidelines](#). It is recommended that this chapter is read before the board design is started.

Note: The ADCs always produce a 16-bit conversion result, and the full 16-bit result is stored in the 16-bit status registers. The 12-bit transfer functions shown in this section correspond to the 12 MSBs in the 16-bit status registers. The four LSBs can be used to minimize quantization effects or improve resolution through averaging or filtering.

Unipolar Mode

Figure 2-1 shows the 12-bit unipolar transfer function for the ADCs. The nominal analog input range to the ADCs is 0V to 1V in this mode. The ADC produces a zero code (000h) when 0V is present on the ADC input and a full scale code of all 1s (FFFh) when 1V is present on the input.

The ADC output coding in unipolar mode is straight binary. The designed code transitions occur at successive integer LSB values such as one LSB, two LSBs, and three LSBs (and so on). The LSB size in volts is equal to $1V/2^{12}$ or $1V/4096 = 244 \mu V$. The analog input channels are differential in nature and require both the positive (V_P) and negative (V_N) inputs of the differential input to be driven. More details on the analog inputs and the kinds of inputs signals that can be accommodated are covered in the [Analog Inputs](#) section.

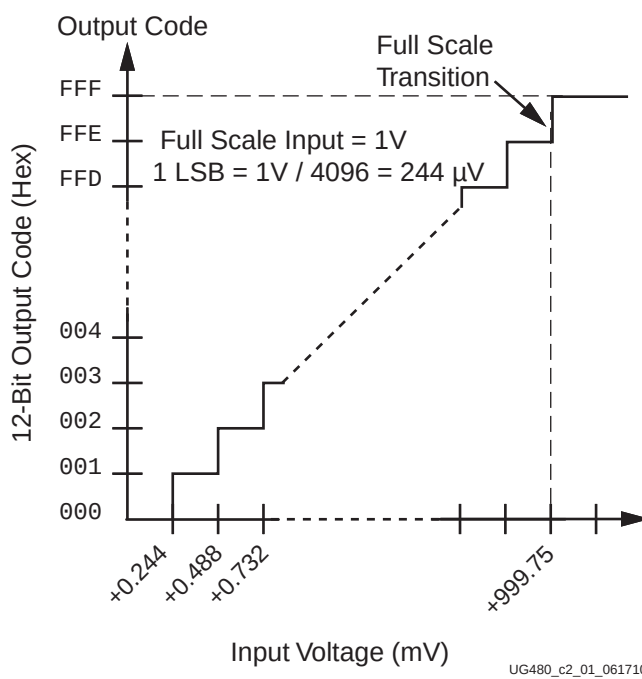


Figure 2-1: Unipolar Transfer Function

Bipolar Mode

When the external analog input channels of the ADCs are configured as bipolar, they can accommodate true differential and bipolar analog signal types (see the [Analog Inputs](#) section). When dealing with differential signal types, it is useful to have both sign and magnitude information about the analog input signal. [Figure 2-2](#) shows the ideal transfer function for bipolar mode operation. The output coding of the ADC in bipolar mode is two's complement and is intended to indicate the sign of the input signal on V_P relative to V_N . The designed code transitions occur at successive integer LSB values, that is, one LSB, two LSBs, three LSBs, etc. The LSB size in volts is equal to $1V/2^{12}$ or $1V/4096 = 244 \mu V$.

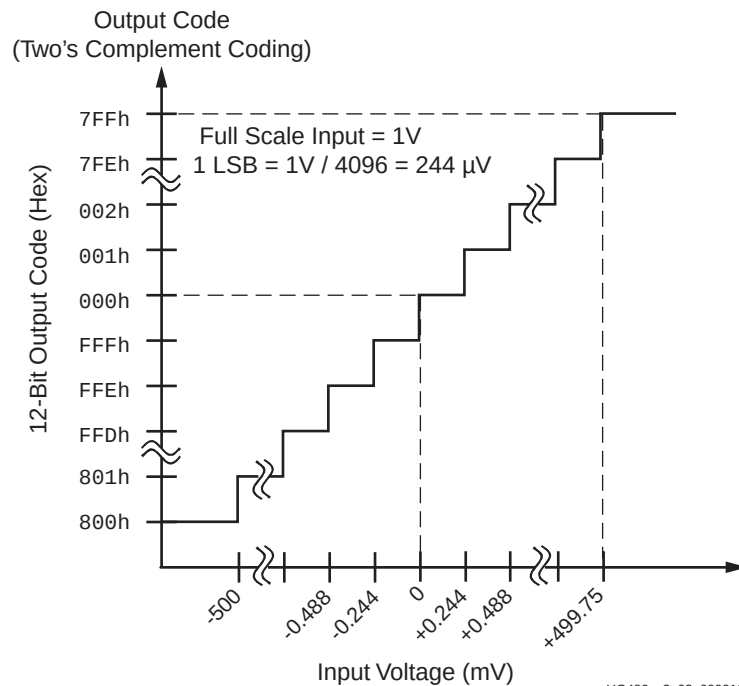
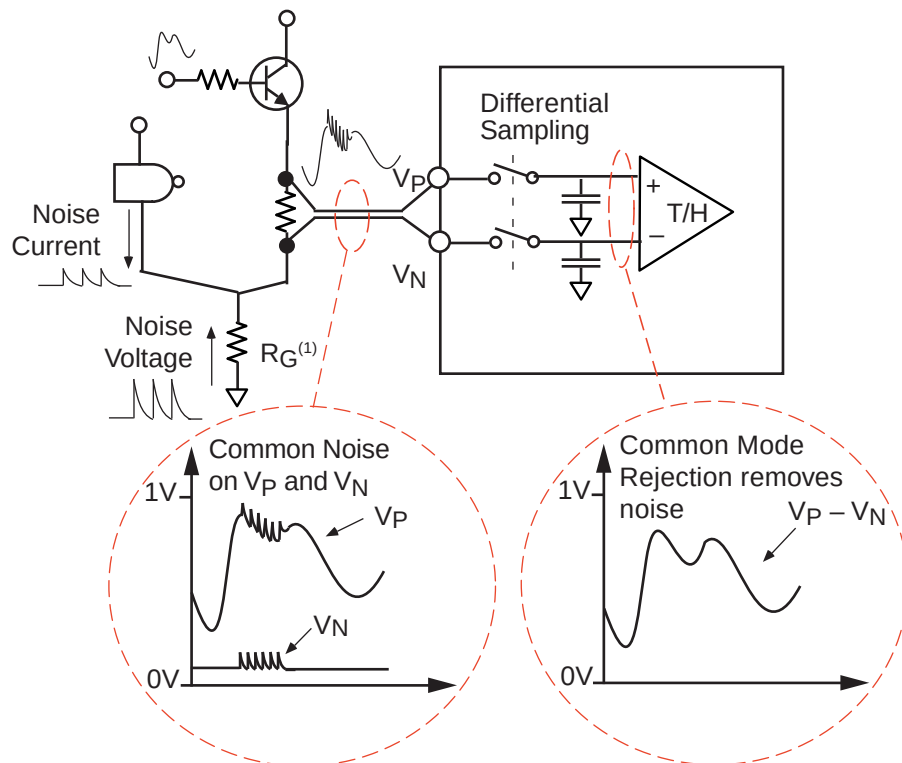


Figure 2-2: Bipolar Transfer Function

Analog Inputs

The analog inputs of the ADC use a differential sampling scheme to reduce the effects of common-mode noise signals. This common-mode rejection improves the ADC performance in noisy digital environments. Figure 2-3 shows the benefits of a differential sampling scheme. Common ground impedances (R_G) easily couple noise voltages (switching digital currents) into other parts of a system. These noise signals can be 100 mV or more. For the ADCs, this noise voltage is equivalent to hundreds of LSBs, thus inducing large measurement errors. The differential sampling scheme samples both the signal and any common mode noise voltages at both analog inputs (V_P and V_N). The common mode signal is effectively subtracted because the Track-and-Hold amplifier captures the difference between V_P and V_N or $V_P - V_N$. To take advantage of the high common mode rejection, users need only connect V_P and V_N in a differential configuration.



Note 1: R_G is Common Ground Impedance.

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Figure 2-3: Common Mode Noise Rejection

Auxiliary Analog Inputs

The auxiliary analog inputs ($VAUXP[15:0]$ and $VAUXN[15:0]$) are analog inputs that are shared with regular digital I/O package balls. The auxiliary analog inputs are automatically enabled when the XADC is instantiated in a design, and these inputs are connected in the instantiation. Only those auxiliary inputs connected in a design are enabled as analog inputs. The XADC auxiliary inputs pins are labeled in the *7 Series FPGA Packaging and Pinout Specification* by appending $_{ADxP}$ and $_{ADxN}$ to the I/O name, where x is the auxiliary pair number. For example, the auxiliary input $VAUXP[15]$ could be designated $IO_{LxxP}_{xx_AD15P}_{xx}$ in the pinout specification.

When designated as analog inputs, these inputs are unavailable for use as digital I/Os. If the I/O is used as a digital I/O, it is subject to the specifications of the configured I/O standard. If the I/O is used as an analog input, the input voltage must adhere to the specifications given in the “Analog-to-Digital Converter” section of the respective 7 series FPGAs data sheet.

It is possible to enable any number of auxiliary analog inputs in an I/O bank and use the remaining as digital I/Os. If there is a mixture of analog and digital I/Os in a bank, the I/O bank must be powered by a supply required to meet the specifications of the digital I/O standard in used. The analog input signal should not exceed the I/O bank supply voltage (V_{CCO}) in this case.

Note: Auxiliary channels 6, 7, 13, 14, and 15 are not supported on Kintex-7 devices.

Adjusting the Acquisition Settling Time

The maximum conversion rate specified for the ADC is 1 MSPS or a conversion time of 1 μ s. In continuous sampling mode (see [Chapter 5, XADC Timing](#)), 26 ADCCLK cycles are required to acquire an analog signal and perform a conversion. This implies a maximum ADCCLK frequency of 26 MHz. If the ACQ (see [Control Registers, page 34](#)) bit has not been set, four ADCCLKs or 150 ns is allowed for the final stages of the acquisition. This “settling” time ensures that the analog input voltage is acquired to a 12-bit accuracy. The settling time can be increased by reducing the ADCCLK frequency or setting the ACQ bit. In the latter case, the settling time is increased to 380 ns (10 ADCCLK cycles), and the conversion rate would be reduced to 812 kSPS for the same ADCCLK frequency. In event timing mode (see [Chapter 5, XADC Timing](#)), the user initiates the conversion cycle by using CONVST or CONVSTCLK, allowing more control over the acquisition time, if required.

Analog Input Description

[Figure 2-4](#) illustrates an equivalent analog input circuit for the external analog input channels in both unipolar and bipolar configurations. The analog inputs consist of a sampling switch and a sampling capacitor used to acquire the analog input signal for conversion. During the ADC acquisition phase, the sampling switch is closed, and the sampling capacitor is charged up to the voltage on the analog input. The time needed to charge this capacitor to its final value (± 0.5 LSBs at 12 bits) is determined by the capacitance of the sampling capacitor (C_{SAMPLE}) and the resistance of the analog multiplexer circuit (R_{MUX}).

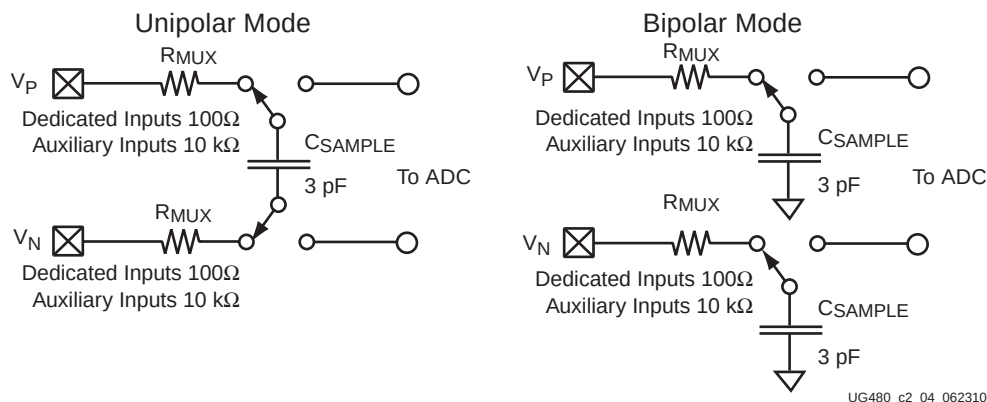


Figure 2-4: Equivalent Analog Input Circuits

The required 12-bit acquisition time is approximated by [Equation 2-1](#).

$$t_{ACQ} = 10 \times R_{MUX} \times C_{SAMPLE} \quad \text{Equation 2-1}$$

This includes enough margin for component variation, etc.

For the dedicated channel (V_P/V_N), the minimum acquisition time required is given by [Equation 2-2](#).

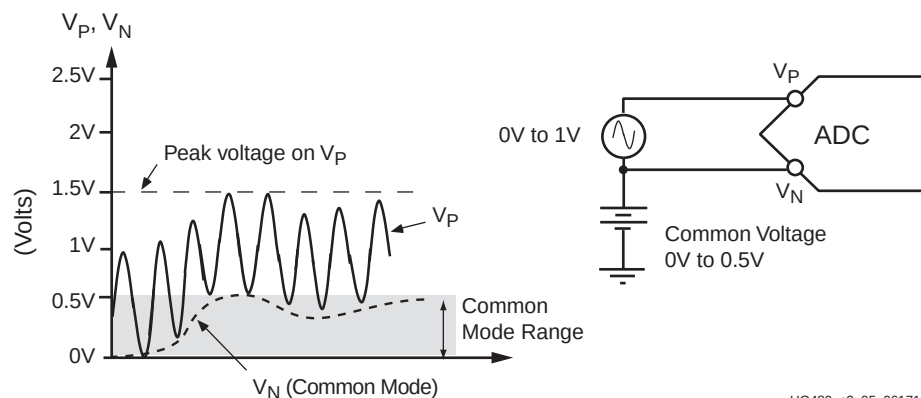
$$t_{ACQ} = 10 \times 100 \times 3 \times 10^{-12} = 3 \text{ ns} \quad \text{Equation 2-2}$$

The auxiliary analog channels (such as, VAUXP[15:0] and VAUXN[15:0]) have a much larger R_{MUX} resistance that is approximately equal to 10 k Ω . [Equation 2-3](#) shows the minimum acquisition time.

$$t_{ACQ} = 10 \times 10 \times 10^3 \times 3 \times 10^{-12} = 300 \text{ ns} \quad \text{Equation 2-3}$$

Unipolar Input Signals

When measuring unipolar analog input signals, the ADCs must operate in a unipolar input mode. This mode is selected by writing to Configuration Register 0 (see [Control Registers, page 34](#)). When unipolar operation is enabled, the differential analog inputs (V_P and V_N) have an input range of 0V to 1.0V. In this mode, the voltage on V_P (measured with respect to V_N) must always be positive. [Figure 2-5](#) shows a typical application of unipolar mode. The V_N input should always be driven by an external analog signal. V_N is typically connected to a local ground or common mode signal. The common mode signal on V_N can vary from 0V to +0.5V (measured with respect to GNDADC). Because the differential input range is from 0V to 1.0V (V_P to V_N), the maximum signal on V_P is 1.5V. [Figure 2-5](#) shows the maximum signal levels on V_N and V_P in unipolar mode, measured with respect to analog ground (GNDADC package ball).



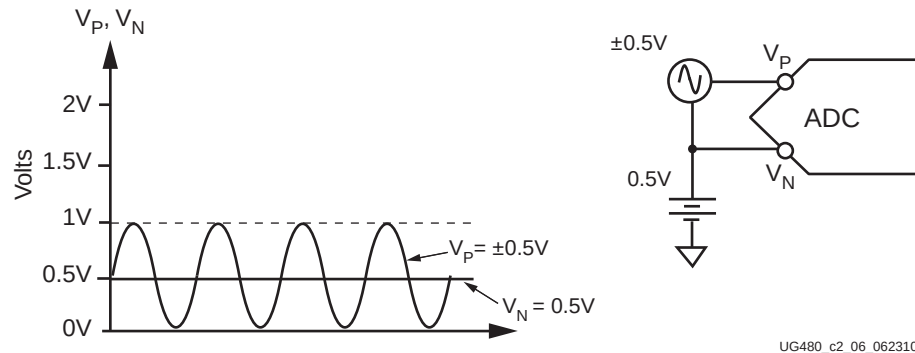
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Figure 2-5: Unipolar Input Signal Range

Bipolar Input Signals

The analog inputs can accommodate analog input signals that are positive and negative with respect to a common mode or reference. To accommodate these types of signals, the analog input must be configured to bipolar mode. Bipolar mode is selected by writing to Configuration Register 0 (see [Control Registers](#)). All input voltages must be positive with respect to analog ground (GNDADC).

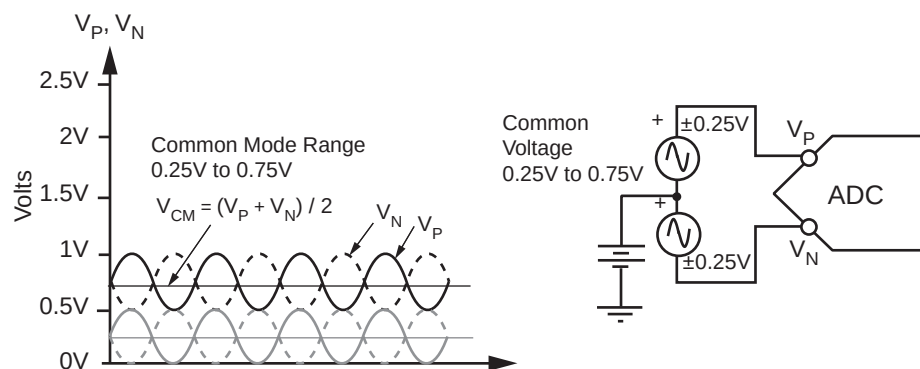
When bipolar operation is enabled, the differential analog input ($V_P - V_N$) can have a maximum input range of $\pm 0.5V$. The common mode or reference voltage should not exceed 0.5V in this case (see [Figure 2-6](#)).



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Figure 2-6: Bipolar Input Signal Range

The bipolar input mode also accommodates inputs signals driven from a true differential source, for example, a balanced bridge. In this case, V_N and V_P can swing positive and negative relative to a common mode or reference voltage (see [Figure 2-7](#)). The maximum differential input ($V_P - V_N$) is $\pm 0.5V$. With maximum differential input voltages of $\pm 0.5V$ and assuming balanced inputs on V_N and V_P , the common mode voltage must lie in the range 0.25V to 0.75V.



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Figure 2-7: Differential Input Signal Range

Temperature Sensor

The XADC contains a temperature sensor that produces a voltage output proportional to the die temperature. Equation 2-4 shows the output voltage of the temperature sensor.

$$Voltage = 10 \times \frac{\kappa T}{q} \times \ln(10) \quad \text{Equation 2-4}$$

Where:

κ = Boltzmann's constant = 1.38×10^{-23} J/K

T = Temperature K (Kelvin) = $^{\circ}\text{C} + 273.15$

q = Charge on an electron = 1.6×10^{-19} C

The output voltage of this sensor is digitized by the ADC to produce a 12-bit digital output code (ADC code). Figure 2-8 illustrates the digital output transfer function for this temperature sensor.

For simplification, the temperature sensor plus the ADC transfer function is rewritten as shown in Equation 2-5.

$$Temperature (^{\circ}\text{C}) = \frac{ADC \text{ Code} \times 503.975}{4096} - 273.15 \quad \text{Equation 2-5}$$

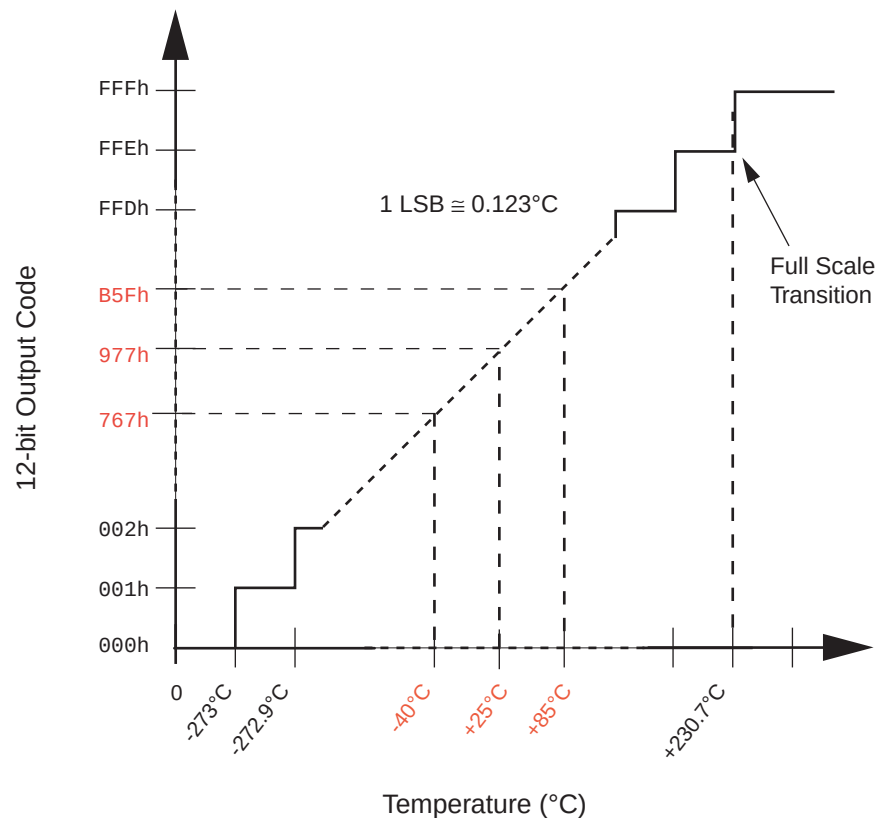


Figure 2-8: Temperature Sensor Transfer Function

The temperature measurement result is stored in the status registers at DRP address 00h. The on-chip temperature sensor has a maximum measurement error of $\pm 4^{\circ}\text{C}$ over a range of -40°C to $+125^{\circ}\text{C}$. Monitoring FPGA on-chip temperature avoids functional and irreversible failures by ensuring critical operating temperatures are not exceeded.

Power Supply Sensor

The XADC also includes on-chip sensors that allow a user to monitor the FPGA power-supply voltages using the ADC. The sensors sample and attenuate (by a factor of three) the power supply voltages V_{CCINT} , V_{CCAUX} , and V_{CCBRAM} on the package power supply balls. Figure 2-9 shows the power-supply sensor transfer function after digitizing by the ADC. The Power Supply sensor can be used to measure voltages in the range 0V to $V_{CCAUX} + 5\%$ with a resolution of approximately 0.73 mV. The transfer function for the supply sensor is shown in Equation 2-6.

$$Voltage = \frac{ADC\ Code}{4096} \times 3V \quad \text{Equation 2-6}$$

The power-supply measurement results for V_{CCINT} , V_{CCAUX} , and V_{CCBRAM} are stored in the status registers at DRP addresses 01h, 02h, and 06h, respectively.

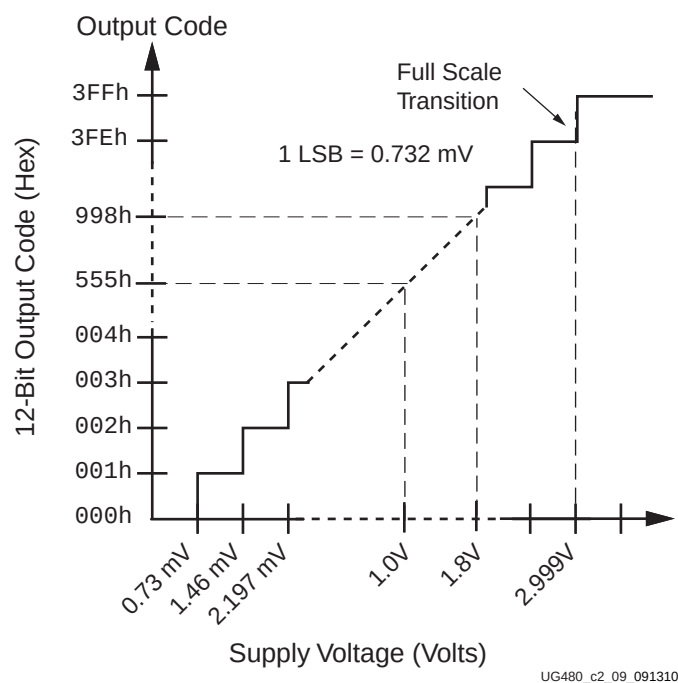


Figure 2-9: Ideal Power Supply Transfer Function

XADC Register Interface

Figure 3-1 illustrates the XADC register interface. All registers in the register interface are accessible through the Dynamic Reconfiguration Port (DRP). The DRP can be accessed via a fabric port or the JTAG TAP. Access is governed by an arbitrator (see [DRP Arbitration](#), page 42). The DRP allows the user to access up to 128 16-bit registers (DADDR[6:0] = 00h to 7Fh). The first 64 access locations (DADDR[6:0] = 00h to 3Fh) are read-only and contain the ADC measurement data. These registers are known as status registers. The control registers are located at addresses 40h to 7Fh and are readable or writable via the DRP. The DRP timing is shown in Figure 5-3.

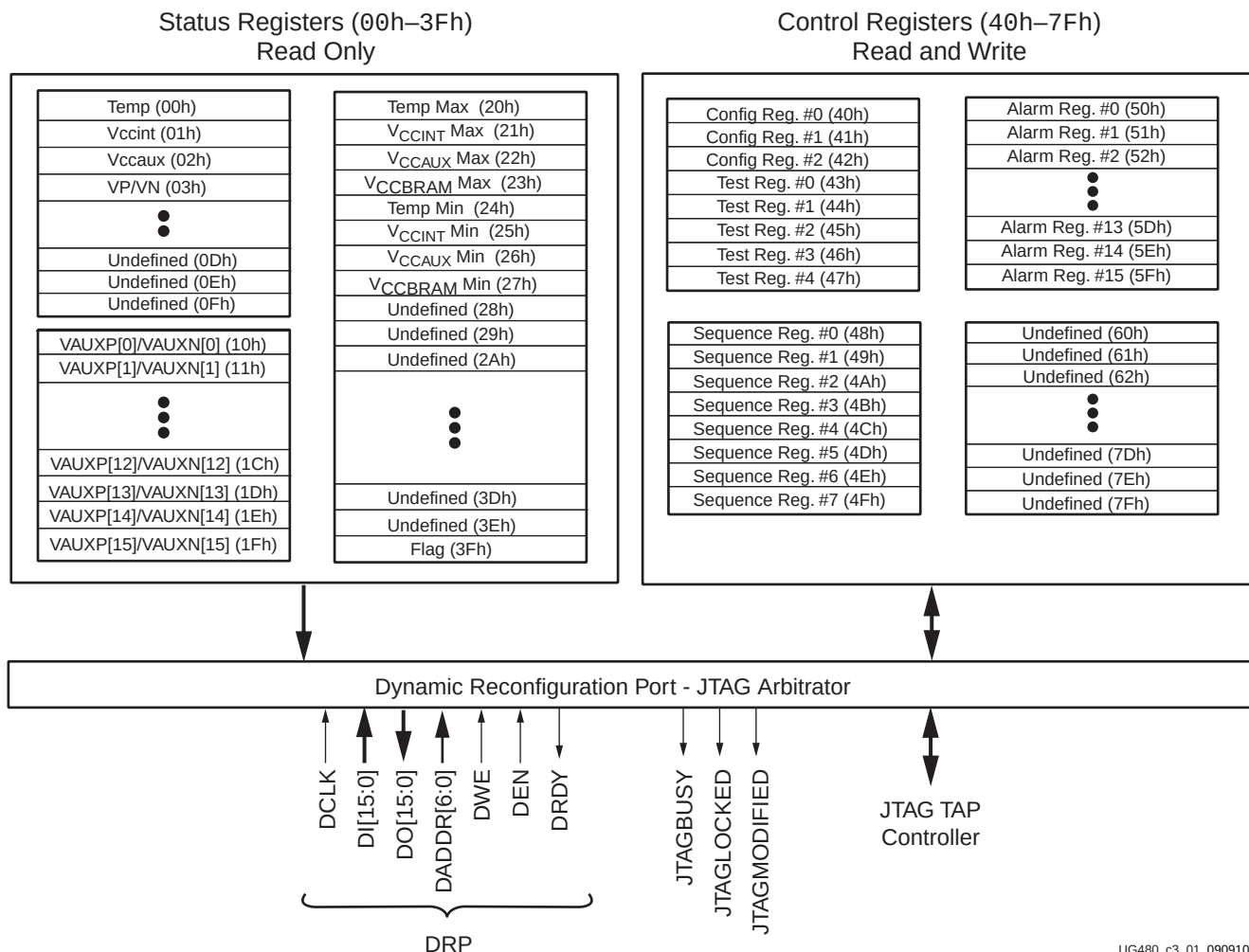


Figure 3-1: XADC Register Interface

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For a detailed description of the DRP timing, refer to the *7 Series FPGAs Configuration User Guide*. For more information on the JTAG DRP interface, see [DRP JTAG Interface](#), page 39.

Status Registers

The first 64 address locations (DADDR[6:0] = 00h to 3Fh) contain the read-only status registers that cannot be initialized when the XADC is instantiated in a design. The status registers contain the results of an analog-to-digital conversion of the on-chip sensors and external channels. All sensors and external analog-input channels have a unique channel address (see [Table 3-7](#), page 37). The measurement result from each channel is stored in a status register with the same address on the DRP.

For example, the result from an analog-to-digital conversion on ADC multiplexer channel 0 (temperature sensor) is stored in the status register at address 00h. The result from ADC multiplexer channel 1 (V_{CCINT}) is stored at address 01h.

The status registers also store the maximum and minimum measurements recorded for the on-chip sensors from the device power-up or the last user reset of the XADC. [Table 3-1](#) defines the status registers.

Table 3-1: Status Registers (Read Only)

Name	Address	Description
Temperature	00h	The result of the on-chip temperature sensor measurement is stored in this location. The data is MSB justified in the 16-bit register. The 12 MSBs correspond to the temperature sensor transfer function shown in Figure 2-8 , page 28.
V_{CCINT}	01h	The result of the on-chip V_{CCINT} supply monitor measurement is stored at this location. The data is MSB justified in the 16-bit register. The 12 MSBs correspond to the supply sensor transfer function shown in Figure 2-9 , page 29.
V_{CCAUX}	02h	The result of the on-chip V_{CCAUX} data supply monitor measurement is stored at this location. The data is MSB justified in the 16 bit register. The 12 MSBs correspond to the supply sensor transfer function shown in Figure 2-9 .
V_P/V_N	03h	The result of a conversion on the dedicated analog input channel is stored in this register. The data is MSB justified in the 16-bit register. The 12 MSBs correspond to the transfer function shown in Figure 2-5 , page 26 or Figure 2-6 , page 27 depending on analog input mode settings.
V_{REFP}	04h	The result of a conversion on the reference input V_{REFP} is stored in this register. The 12 MSBs correspond to the ADC transfer function shown in Figure 2-9 . The data is MSB justified in the 16-bit register. The supply sensor is used when measuring V_{REFP} .
V_{REFN}	05h	The result of a conversion on the reference input V_{REFN} is stored in this register. The 12 MSBs correspond to the ADC transfer function shown in Figure 2-9 . The data is MSB justified in the 16-bit register. The supply sensor is used when measuring V_{REFN} .

Table 3-1: Status Registers (Read Only) (Cont'd)

Name	Address	Description
V _{CC} BRAM	06h	The result of the on-chip V _{CC} BRAM supply monitor measurement is stored at this location. The data is MSB justified in the 16-bit register. The 12 MSBs correspond to the supply sensor transfer function shown in Figure 2-9.
Undefined	07h	This location is unused and contains invalid data.
Supply A Offset	08h	The calibration coefficient for the supply sensor offset using ADC A is stored at this location.
ADC A Offset	09h	The calibration coefficient for the ADC A offset is stored at this location.
ADC A Gain	0Ah	The calibration coefficient for the ADC A gain error is stored at this location.
Undefined	0Bh to 0Fh	These locations are unused and contain invalid data.
VAUXP[15:0]/ VAUXN[15:0]	10h to 1Fh	The results of the conversions on auxiliary analog input channels are stored in this register. The data is MSB justified in the 16-bit register. The 12 MSBs correspond to the transfer function shown in Figure 2-5 or Figure 2-6 depending on analog input mode settings.
Max Temp	20h	Maximum temperature measurement recorded since power-up or the last XADC reset.
Max V _{CC} INT	21h	Maximum V _{CC} INT measurement recorded since power-up or the last XADC reset.
Max V _{CC} AUX	22h	Maximum V _{CC} AUX measurement recorded since power-up or the last XADC reset.
Max V _{CC} BRAM	23h	Maximum V _{CC} BRAM measurement recorded since power-up or the last XADC reset.
Min Temp	24h	Minimum temperature measurement recorded since power-up or the last XADC reset.
Min V _{CC} INT	25h	Minimum V _{CC} INT measurement recorded since power-up or the last XADC reset.
Min V _{CC} AUX	26h	Minimum V _{CC} AUX measurement recorded since power-up or the last XADC reset.
Min V _{CC} BRAM	27h	Minimum V _{CC} BRAM measurement recorded since power-up or the last XADC reset.
Undefined	28h to 2Fh	These locations are unused and contain invalid data.
Supply B Offset	30h	The calibration coefficient for the supply sensor offset using ADC B is stored at this location.
ADC B Offset	31h	The calibration coefficient for the ADC B offset is stored at this location.
ADC B Gain	32h	The calibration coefficient for the ADC B gain error is stored at this location.

Table 3-1: Status Registers (Read Only) (Cont'd)

Name	Address	Description
Undefined	33h to 3Eh	These locations are unused and contain invalid data.
Flag	3Fh	This register contains general status information (see Flag Register).

Flag Register

The Flag Register is shown in [Figure 3-2](#). The bit definitions are described in [Table 3-2](#).

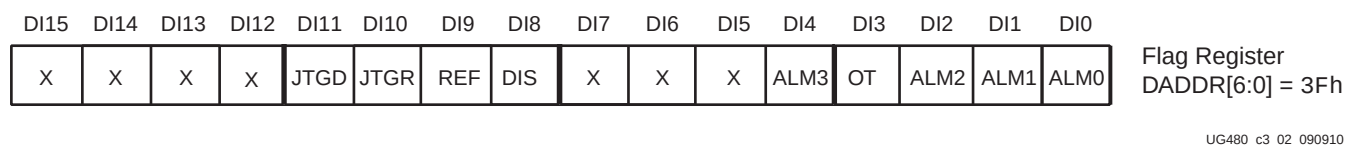


Figure 3-2: Flag Registers

Table 3-2: Flag Register Bit Definitions

Bit	Name	Description
DI4, DI2, DI1, DI0	ALM3 to ALM0	These bits reflect the status of the alarm output ALM[3:0].
DI3	OT	This bit reflects the status of the Over Temperature logic output.
DI8	DIS	When this bit is a logic 1, the XADC is disabled by connecting the supplies and reference inputs to AGND.
DI9	REF	When this bit is a logic 1, the ADC is using the internal voltage reference. When this bit is a logic 0, the external reference is being used.
DI10	JTGR	A logic 1 indicates that the JTAG_XADC BitGen option has been used to restrict JTAG access to read only. See DRP JTAG Interface for more information.
DI11	JTGD	A logic 1 indicates that the JTAG_XADC BitGen option has been used to disable all JTAG access. See DRP JTAG Interface for more information.

Control Registers

The XADC has 32 control registers that are located at DRP addresses 40h to 5Fh (see [Table 3-3](#)). These registers are used to configure the XADC operation. All XADC functionality is controlled through these registers.

These control registers are initialized using the XADC attributes when the XADC is instantiated in a design. This means that the XADC can be configured to start in a predefined mode after FPGA configuration.

Table 3-3: XADC Control Registers

Name	Address	Software Attribute	Description
Configuration Register 0	40h	INIT_40	These are XADC configuration registers (see Configuration Registers (40h to 42h)).
Configuration Register 1	41h	INIT_41	
Configuration Register 2	42h	INIT_42	
Test Registers 0 to 4	43h to 47h	INIT_43 to INIT_47	These are test registers. The default initialization is 0000h. These registers are used for factory test and should be left at the default initialization.
Sequence Registers	48h to 4Fh	INIT_48 to INIT_4F	These registers are used to program the channel sequencer function (see Chapter 4, XADC Operating Modes).
Alarm Registers	50h to 5Fh	INIT_50 to INIT_5F	These are the alarm threshold registers for the XADC alarm function (see Automatic Alarms, page 55).

Configuration Registers (40h to 42h)

The first three registers in the control register block configure the XADC operating modes. These registers are known as XADC configuration registers. Their bit definitions are illustrated in [Figure 3-3](#).

Note: Bits shown as 0 should always be left set to 0.

The configuration registers can be modified through the DRP after the FPGA has been configured. For example, a soft microprocessor or state machine can be used to alter the contents of the XADC control registers at any time during normal operation. [Table 3-4](#) through [Table 3-6](#) define the bits for the three configuration registers.

DI15	DI14	DI13	DI12	DI11	DI10	DI9	DI8	DI7	DI6	DI5	DI4	DI3	DI2	DI1	DI0	Config Reg #0 DADDR[6:0] = 40h
CAVG	0	AVG1	AVG0	MUX	B $\bar{U}$	E $\bar{C}$	ACQ	0	0	0	CH4	CH3	CH2	CH1	CH0	
DI15	DI14	DI13	DI12	DI11	DI10	DI9	DI8	DI7	DI6	DI5	DI4	DI3	DI2	DI1	DI0	Config Reg #1 DADDR[6:0] = 41h
SEQ3	SEQ2	SEQ1	SEQ0	0	0	0	ALM3	CAL3	CAL2	CAL1	CAL0	ALM2	ALM1	ALM0	OT	
DI15	DI14	DI13	DI12	DI11	DI10	DI9	DI8	DI7	DI6	DI5	DI4	DI3	DI2	DI1	DI0	Config Reg #2 DADDR[6:0] = 42h
CD7	CD6	CD5	CD4	CD3	CD2	CD1	CD0	0	0	PD1	PD0	0	0	0	0	

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Figure 3-3: Configuration Registers Bit Definitions

Table 3-4: Configuration Register 0 Bit Definitions

Bit	Name	Description
DI0 to DI3	CH0 to CH4	When operating in single channel mode or external multiplexer mode, these bits are used to select the ADC input channel. See Table 3-7 for the channel assignments.
DI8	ACQ	This bit is used to increase the settling time available on external analog inputs in continuous sampling mode by six ADCCLK cycles (see Chapter 2, Analog-to-Digital Converter and Chapter 5, XADC Timing). The acquisition time is increased by setting this bit to logic 1.
DI9	$\overline{EC}$	This bit is used to select either continuous or event-driven sampling mode for the ADC (see Chapter 5, XADC Timing). A logic 1 places the ADC in event-driven sampling mode and a logic 0 places the ADC in continuous sampling mode.
DI10	$\overline{BU}$	This bit is used in single channel mode to select either unipolar or bipolar operating mode for the ADC analog inputs (see Analog Inputs, page 24). A logic 1 places the ADC in bipolar mode and a logic 0 places the ADC in unipolar mode.
DI11	MUX	This bit should be set to a logic 1 to enable external multiplexer mode. See External Multiplexer Mode for more information.
DI12, DI13	AVG0, AVG1	These bits are used to set the amount of sample averaging on selected channels in both single channel and sequence modes (see Table 3-8 and Chapter 4, XADC Operating Modes).
DI15	CAVG	This bit is used to disable averaging for the calculation of the calibration coefficients. Averaging is enabled by default (logic 0). To disable averaging, set this bit to logic 1. Averaging is fixed at 16 samples.

Table 3-5: Configuration Register 1 Bit Definitions

Bit	Name	Description
DI0	OT	This bit is used to disable the over-temperature signal. The alarm is disabled by setting this bit to logic 1.
DI1 to DI3, DI8	ALM0 to ALM3	These bits are used to disable individual alarm outputs for temperature, V_{CCINT} , V_{CCAUX} , and V_{CCBRAM} , respectively. A logic 1 disables an alarm output.
DI4 to DI7	CAL0 to CAL3	These bits enable the application of the calibration coefficients to the ADC and on-chip supply sensor measurements. A logic 1 enables calibration and a logic 0 disables calibration. For bit assignments, see Table 3-10 .
DI12 to DI15	SEQ0 to SEQ3	These bits enable the channel-sequencer function. For the bit assignments, see Table 3-9 . See Chapter 4, XADC Operating Modes , for more information.

Table 3-6: Configuration Register 2 Bit Definitions

Bit	Name	Description
DI4, DI5	PD0, PD1	Power-down bits for the XADC. The entire XADC block can be powered down by setting PD1 = PD0 = 1. ADC B can also be powered down by setting PD1 = 1 and PD0 = 0. See Table 3-11.
DI8 to DI15	CD0 to CD7	These bits select the division ratio between the DRP clock (DCLK) and the lower frequency ADC clock (ADCCLK) used for the ADC (see Chapter 5, XADC Timing). For bit assignments, see Table 3-12.

Table 3-7: ADC Channel Select

ADC Channel	CH4	CH3	CH2	CH1	CH0	Description
0	0	0	0	0	0	On-chip temperature
1	0	0	0	0	1	Average on-chip V_{CCINT}
2	0	0	0	1	0	Average on-chip V_{CCAUX}
3	0	0	0	1	1	V_P , V_N - Dedicated analog inputs
4	0	0	1	0	0	V_{REFP} (1.25V) ⁽¹⁾
5	0	0	1	0	1	V_{REFN} (0V) ⁽¹⁾
6	0	0	1	1	0	Average on-chip V_{CCBRAM}
7	0	0	1	1	1	Invalid channel selection
8	0	1	0	0	0	Carry out an XADC calibration
9–15	...	...	...	...	...	Invalid channel selection
16	1	0	0	0	0	VAUXP[0], VAUXN[0] - Auxiliary channel 0
17	1	0	0	0	1	VAUXP[1], VAUXN[1] - Auxiliary channel 1
18–31	...	...	...	...	...	VAUXP[2:15], VAUXN[2:15] - Auxiliary channels 2 to 15 ⁽²⁾

Notes:

- These channel selection options are used for XADC self-check and calibration operations. When these channels are selected, the supply sensor is connected to V_{REFP} or V_{REFN} .
- Auxiliary channels 6, 7, 13, 14, and 15 are not supported on Kintex™-7 devices.

Table 3-8: Averaging Filter Settings

AVG1	AVG0	Function
0	0	No averaging
0	1	Average 16 samples
1	0	Average 64 samples
1	1	Average 256 samples

Table 3-9: Sequencer Operation Settings

SEQ3	SEQ2	SEQ1	SEQ0	Function
0	0	0	0	Default Mode
0	0	0	1	Single pass sequence
0	0	1	0	Continuous sequence mode
0	0	1	1	Single Channel mode (Sequencer Off)
0	1	X	X	Simultaneous Sampling Mode
1	0	X	X	Independent ADC Mode
1	1	X	X	Default Mode

Table 3-10: Calibration Enables

Name	Description
CAL0	ADCs offset correction enable
CAL1	ADCs offset and gain correction enable
CAL2	Supply Sensor offset correction enable
CAL3	Supply Sensor offset and gain correction enable

Table 3-11: Power Down Selection

PD1	PD0	Description
0	0	Default. All XADC blocks powered up
0	1	Not valid - do not select
1	0	ADC B powered down
1	1	XADC powered down

Table 3-12: DCLK Division Selection⁽¹⁾

CD7	CD6	CD5	CD4	CD3	CD2	CD1	CD0	Division
0	0	0	0	0	0	0	0	2
0	0	0	0	0	0	0	1	2
0	0	0	0	0	0	1	0	2
0	0	0	0	0	0	1	1	3
0	0	0	0	0	1	0	0	4
.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.
1	1	1	1	1	1	1	0	254
1	1	1	1	1	1	1	1	255

Notes:

1. Minimum division ratio is 2, for example, $ADCCLK = DCLK/2$.

Test Registers (43h to 47h)

These registers, intended for factory test purposes only, have a default status of zero. The user must not write to these registers.

Channel Sequencer Registers (48h to 4Fh)

These registers are used to program the channel sequencer functionality. For more information see [Automatic Channel Sequencer](#), page 45.

Alarm Registers (50h to 57h)

These registers are used to program the alarm thresholds for the automatic alarms on the internally monitored channels, temperature, V_{CCINT} , V_{CCAUX} , and V_{CCBRAM} . For more information, see [Automatic Alarms](#), page 55.

DRP JTAG Interface

The XADC uses a full JTAG interface extension to the DRP interface. This allows read/write access to the XADC DRP through the existing on-chip JTAG infrastructure. No instantiation is required to access the DRP interface over JTAG. A Boundary-Scan instruction (6-bit instruction = 110111) called XADC_DRP, added to 7 series FPGAs, allows access to the DRP through the JTAG TAP. All XADC JTAG instructions are 32 bits wide. For more information on the 7 series FPGA boundary-scan instructions and usage, see the *7 Series FPGAs Configuration User Guide*. Read and Write operations using the XADC JTAG DRP interface are described in the next sections. Users unfamiliar with basic JTAG functionality should become familiar with the JTAG standard (IEEE 1149.1) before proceeding.

Note: JTAG access can be limited to read only or completely disabled by using the **JTAG_XADC : [Enable | Disable | Status_only]** BitGen option.

XADC DRP JTAG Write Operation

[Figure 3-4](#) shows a timing diagram for a write operation to the XADC DRP through the JTAG TAP. The DRP is accessed through the XADC Data Register (XADC DR). Before the XADC DR is accessed, the instruction register (IR) must first be loaded with the XADC instruction. The controller is placed in the IR-scan mode, and the XADC instruction is shifted to the IR.

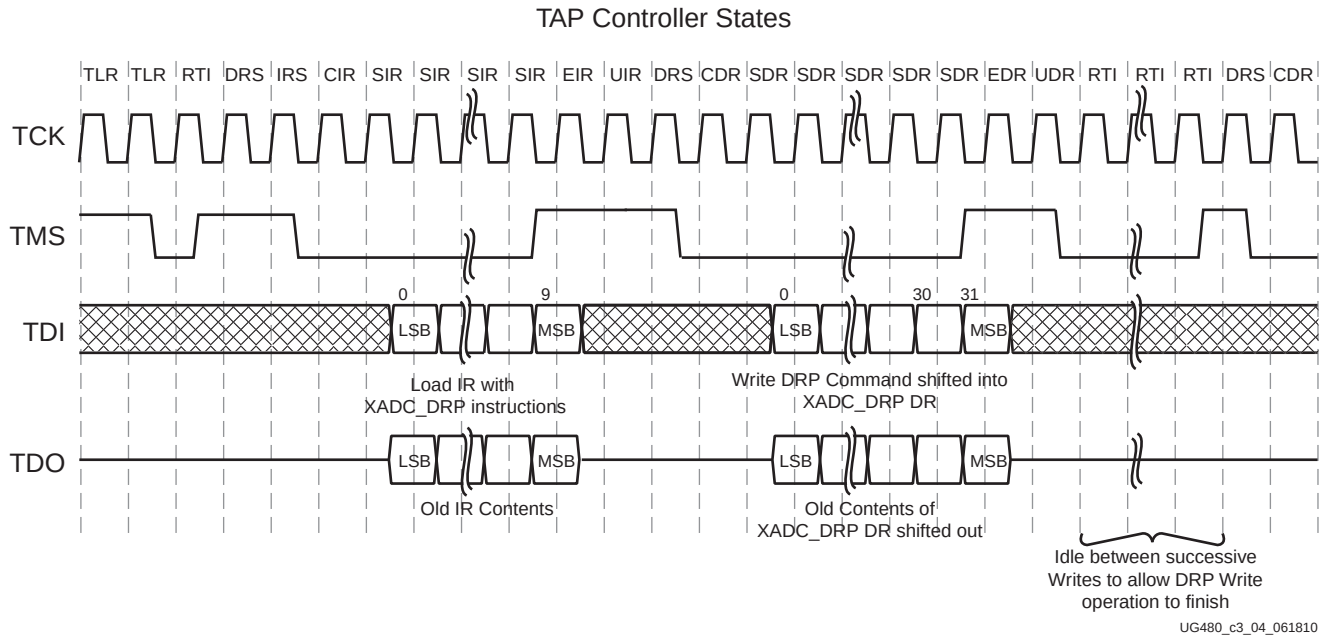


Figure 3-4: XADC JTAG DRP Write

After the XADC instruction is loaded, all data register (DR)-scan operations are carried out on the XADC DR. When the data shifted into XADC DR is a JTAG DRP Write command, the XADC DRP arbitrator carries out a DRP write. The format of this write command is described in [JTAG DRP Commands](#). The XADC DR contents are transferred to the XADC DRP arbitrator (see [DRP Arbitration](#)) during the Update-DR state. After the Update-DR state, the arbitrator manages the new data transfer to the XADC DRP register. This takes up to six DRP Clocks (DCLK) cycles if a DRP access from the FPGA logic is already in progress.

During the Capture-DR phase (just before data is shifted into the XADC DR), DRP data is captured from the arbitrator. Depending on the last JTAG DRP command, this data can be old data, previously written to the DRP, or requested new read data (see [XADC DRP JTAG Read Operation](#)). This captured data is shifted out (LSB first) on DO as the new JTAG DRP command is shifted in. The 16 LSBs of this 32-bit word contain the JTAG DRP data. The 16 MSBs are set to zero.

If multiple writes to the XADC DR are taking place, it might be necessary to idle the TAP controller for several TCK cycles before advancing to the next write operation (see [Figure 3-4](#)). The idle cycles allow the arbitrator to complete the write operation to the XADC DRP register. If DCLK is running approximately $6 \times \text{TCK}$, these idle states are not necessary. However, inserting 10 or so idle states ensures all transfers are inherently safe.

XADC DRP JTAG Read Operation

[Figure 3-5](#) shows the timing for an XADC DR Read operation. The IR should contain the DR-scan operation (XADC_DRP instruction). A JTAG read from the XADC DRP is a two-step operation.

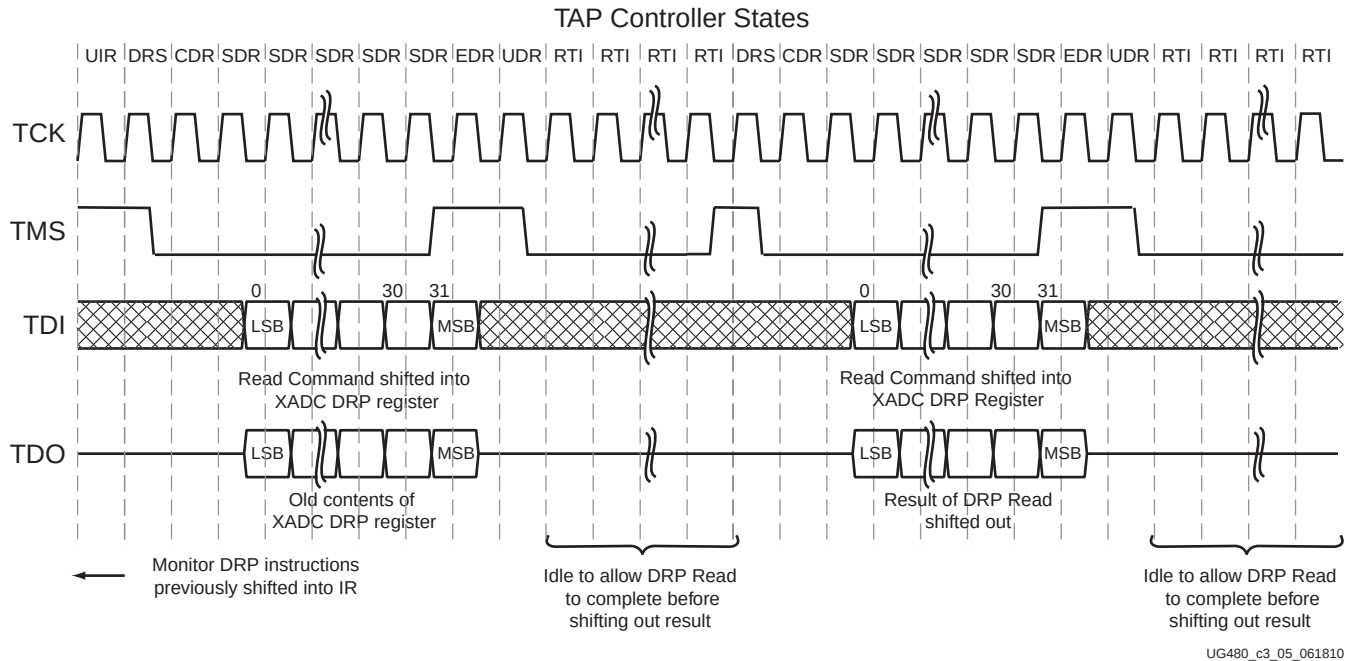


Figure 3-5: XADC JTAG DRP Read

First, the XADC DR is loaded with the Read DRP instruction. This instruction is transferred to the arbitrator during the Update-DR state. Then the arbitrator reads the selected DRP register and stores the newly read 16-bit data. This operation takes several DCLK cycles to complete.

During the DR-Capture phase of the next DR-scan operation, newly read data is transferred from the arbitrator to the XADC DR. This 16-bit data (stored in the 16 LSBs of the 32-bit word) is then shifted out on TDO during the subsequent shift operation (see [Figure 3-5](#)). The timing diagram shows a number of idle states at the end of the first DR-scan operation, allowing the arbitrator enough time to fetch the XADC DRP data.

As mentioned previously, if the DCLK frequency is significantly faster than the TCK, these idle states might not be required.

Implementing a DR-scan operation before the arbitrator has completed the DRP-read operation results in old DRP data being transferred to the XADC DR during the DR-capture phase.

To ensure reliable operation over all operating clock frequencies, a minimum of 10 Run-Test-Idle (RTI) states should be inserted. Multiple read operations can be pipelined, as shown in [Figure 3-5](#). Thus, as the result of a read operation is being shifted out of the XADC DR, an instruction for the next read can be shifted in.

JTAG DRP Commands

The data shifted into the 32-bit XADC DR during a DR-scan operation instructs the arbitrator to carry out a write, read, or no operation on the XADC DRP. [Figure 3-6](#) shows the data format of the JTAG DRP command loaded into the XADC DR. The first 16 LSBs of XADC DR [15:0] contain the DRP register data. For both read and write operations, the address bits XADC DR [25:16] hold the DRP target register address. The command bits XADC DR [29:26] specify a read, write, or no operation (see [Table 3-13](#)).

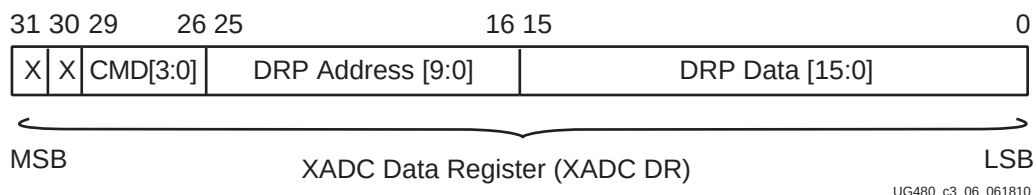


Figure 3-6: XADC JTAG DRP Command

Table 3-13: JTAG DRP Commands

CMD[3:0]				Operation
0	0	0	0	No operation
0	0	0	1	DRP Read
0	0	1	0	DRP Write
-	-	-	-	Not defined

It is also possible to enable the auxiliary analog input channel preconfiguration of the FPGA, allowing external analog voltages (on the PCB) to be monitored using the JTAG TAP before configuration. The auxiliary channels are enabled by writing 0001h to DRP address 02h. This address lies within the read-only status register address space and normally holds the result of a V_{CCAUX} measurement. However, a write to this address enables the auxiliary inputs. This function only works prior to configuration. After configuration, these inputs must be explicitly instantiated in the design.

DRP Arbitration

Because the DRP registers are accessed from two different ports (interconnect and JTAG TAP), access must be carefully managed. An arbitrator is implemented to manage potential conflicts between the FPGA logic (fabric) and JTAG port. Arbitration is managed on a per transaction basis (a transaction is a single read/write operation to the DRP). The arbitration rules are as follows:

1. A JTAG transaction cannot be interrupted by the FPGA logic (fabric). The fabric transaction is queued by the arbitrator until the JTAG transaction has finished, and then the fabric transaction is completed.
2. A JTAG transaction cannot interrupt a fabric transaction already in progress. As soon as the fabric transaction is finished, then the JTAG transaction is completed.

Three status signals help manage access through the interconnect when the JTAG port is also being used: JTAGBUSY, JTAGMODIFIED, and JTAGLOCKED.

JTAGBUSY

JTAGBUSY becomes active during the update phase of a DRP transaction through the JTAG TAP. This signal resets when the JTAG XADC DR transaction is completed. Each read/write to the XADC DR is treated as an individual transaction. If DRP access initiates through the interconnect port when JTAGBUSY is High, then the arbitrator queues this request for a read/write through the FPGA logic. DRDY does not go active until JTAGBUSY transitions Low and the interconnect transaction is completed. A second DRP access through the FPGA logic must not be initiated until the DRDY for the initial access becomes active and indicates the read/write was successful. If an interconnect access is in progress when a JTAG DRP transaction initiates, the interconnect access is completed before the JTAG transaction.

JTAGMODIFIED

Whenever there is a JTAG write (JTAG reads typically occur more often) to any register in the DRP, the application (FPGA) must be notified about the potential change of configuration. Thus, the JTAGMODIFIED signal transitions High after a JTAG write. A subsequent DRP read/write resets the signal.

JTAGLOCKED

In some cases, it is simpler for the JTAG user to take DRP ownership for a period by locking out access through the interconnect. This is useful in a diagnostic situation where a large number of DRP registers are modified through the JTAG TAP. When a JTAGLOCKED request is made, the JTAGLOCKED signal transitions to the active High state. The signal remains High until the port is unlocked again. No read or write access is possible via the DRP fabric port when the JTAGLOCKED signal is High. The JTAGLOCKED signal is activated by writing 0001h to DRP address 00h. The JTAGLOCKED signal is reset by writing 0000h to DRP address 00h.

JTAGLOCKED is also used to indicate when the DRP is ready for a read or write when the DCLK is first connected or when DCLK becomes active again after a period of inactivity. It can take up to 10 DCLK cycles for JTAGLOCKED to go Low after DCLK becomes active.

Note: The XADC automatically switches over to an on-chip clock oscillator if a missing DCLK is detected.

XADC JTAG Reset

A user reset of the XADC can also be initiated using the JTAG interface. The XADC is reset by writing xxxh (any 16-bit value) to DRP address 03h. The JTAG reset has the same effect as pulsing the RESET pin.

XADC Operating Modes

The XADC includes a number of operating modes that cover some of the most common use cases for this kind of functionality. The most basic mode of operation is called *default mode*, where the XADC monitors all on-chip sensors and requires no configuration of the XADC. In the *simultaneous sampling mode*, the sequencer is used to operate both ADCs in parallel to sample two external analog inputs and store results in the status registers.

Single Channel Mode

Single channel mode is the most basic way users can modify the operation of the XADC. This mode is enabled when bits SEQ3 to SEQ0 in Control Register 41h are set to 0011 (see [Table 3-9, page 38](#)). In this mode, the user must select the channel for analog-to-digital conversion by writing to bit locations CH4 to CH0 in Control Register 40h. Various configurations for single channel mode, such as analog input mode ($\overline{BU}$) and settling time (ACQ), must also be set by writing to Control Register 40h. In applications where many channels need to be monitored, there can be a significant overhead for the microprocessor or other controller. To automate this task, a function called the automatic channel sequencer is provided.

Automatic Channel Sequencer

The automatic channel sequencer sets up a range of predefined operating modes, where a number of channels (on-chip sensors and external inputs) are used. The sequencer automatically selects the next channel for conversion, sets the averaging, configures the analog input channels, sets the required settling time for acquisition, and stores the results in the status registers based on a once off setting. The sequencer modes are set by writing to the SEQ3, SEQ2, SEQ1, and SEQ0 bits in Configuration Register 1 (see [Table 3-9](#)).

The channel sequencer functionality is implemented using eight control registers from addresses 48h to 4Fh on the DRP (see [Control Registers, page 34](#)). These eight registers can be viewed as four pairs of 16-bit registers. Each pair of registers controls one aspect of the sequencer functionality. Individual bits in each pair of registers (32 bits) enable specific functionality for a particular ADC channel. The four pairs of registers are:

- [ADC Channel Selection Registers \(48h and 49h\)](#)
- [ADC Channel Averaging \(4Ah and 4Bh\)](#)
- [ADC Channel Analog-Input Mode \(4Ch and 4Dh\)](#)
- [ADC Channel Settling Time \(4Eh and 4Fh\)](#)

ADC Channel Selection Registers (48h and 49h)

The ADC channel selection registers enable and disable a channel in the automatic channel sequencer. The bits for these registers are defined in [Table 4-1](#) and [Table 4-2](#). The two 16-bit registers are used to enable or disable the associated channels. A logic 1 enables a particular channel in the sequence. The sequence order is fixed starting from the LSB (bit 0) of register 48h and ending with the MSB (bit 15) of register 49h. The behavior of the channel sequencer does change for simultaneous sampling mode and independent ADC mode (see [Simultaneous Sampling Mode](#) and [Independent ADC Mode](#), respectively).

Table 4-1: Sequencer On-Chip Channel Selection (48h)

Sequence Number	Bit	ADC Channel	Description
1	0	8	XADC calibration
-	1	9	Invalid channel selection
	2	10	
	3	11	
	4	12	
	5	13	
	6	14	
	7	15	
2	8	0	On-chip temperature
3	9	1	Average on-chip V_{CCINT}
4	10	2	Average on-chip V_{CCAUX}
5	11	3	V_P , V_N - Dedicated analog inputs
6	12	4	V_{REFP} (1.25V)
7	13	5	V_{REFN} (0V)
8	14	6	Average on-chip V_{CCBRAM}
-	15	7	Invalid channel selection

Table 4-2: Sequencer Auxiliary Channel Selection (49h)

Sequence Number	Bit	ADC Channel	Description
9	0	16	VAUXP[0], VAUXN[0] - Auxiliary channel 0
10	1	17	VAUXP[1], VAUXN[1] - Auxiliary channel 1
11	2	18	VAUXP[2], VAUXN[2] - Auxiliary channel 2
12	3	19	VAUXP[3], VAUXN[3] - Auxiliary channel 3
13	4	20	VAUXP[4], VAUXN[4] - Auxiliary channel 4
14	5	21	VAUXP[5], VAUXN[5] - Auxiliary channel 5

Table 4-2: Sequencer Auxiliary Channel Selection (49h) (Cont'd)

Sequence Number	Bit	ADC Channel	Description
15	6	22	VAUXP[6], VAUXN[6] - Auxiliary channel 6 ⁽¹⁾
16	7	23	VAUXP[7], VAUXN[7] - Auxiliary channel 7 ⁽¹⁾
17	8	24	VAUXP[8], VAUXN[8] - Auxiliary channel 8
18	9	25	VAUXP[9], VAUXN[9] - Auxiliary channel 9
19	10	26	VAUXP[10], VAUXN[10] - Auxiliary channel 10
20	11	27	VAUXP[11], VAUXN[11] - Auxiliary channel 11
21	12	28	VAUXP[12], VAUXN[12] - Auxiliary channel 12
22	13	29	VAUXP[13], VAUXN[13] - Auxiliary channel 13 ⁽¹⁾
23	14	30	VAUXP[14], VAUXN[14] - Auxiliary channel 14 ⁽¹⁾
24	15	31	VAUXP[15], VAUXN[15] - Auxiliary channel 15 ⁽¹⁾

Notes:

1. Auxiliary channels 6, 7, 13, 14, and 15 are not supported on Kintex-7 devices.

ADC Channel Averaging (4Ah and 4Bh)

The ADC channel averaging registers enable and disable the averaging of the channel data in a sequence. The result of a measurement on an averaged channel is generated by using 16, 64, or 256 samples. The amount of averaging is selected by using the AVG1 and AVG0 bits in Configuration Register 0 (see [Control Registers, page 34](#)). These registers also have the same bit assignments as the channel sequence registers listed in [Table 4-1](#) and [Table 4-2](#).

Averaging can be selected independently for each channel in the sequence. When averaging is enabled for some of the channels of the sequence, the EOS is only pulsed after the sequence has completed the amount of averaging selected by using AVG1 and AVG0 bits (see [Table 3-8, page 37](#)). If a channel in the sequence does not have averaging enabled, its status register is updated for every pass through the sequencer. When a channel has averaging enabled, its status register is only updated after the averaging is complete. An example sequence is Temperature and $V_{AUX}[1]$, where an averaging of 16 is enabled on $V_{AUX}[1]$. The sequence is Temperature, $V_{AUX}[1]$, Temperature, $V_{AUX}[1]$, ..., Temperature, $V_{AUX}[1]$ for each of the conversions where the temperature status register is updated. The $V_{AUX}[1]$ status register is updated after the averaging of the 16 conversions.

If averaging is enabled for the calibration channel by setting CAVG to a logic 0 (see [Control Registers, page 34](#)), the coefficients are updated after the first pass through the sequence. Subsequent updates to coefficient registers require 16 conversions before the coefficients are updated. Averaging is fixed at 16 samples for calibration.

ADC Channel Analog-Input Mode (4Ch and 4Dh)

These registers are used to configure an ADC channel as either unipolar or bipolar in the automatic sequence (see [Analog Inputs, page 24](#)). These registers also have the same bit assignments as the channel sequence registers listed in [Table 4-1](#) and [Table 4-2](#). However, only external analog input channels, such as the dedicated input channels (V_P and V_N) and the auxiliary analog inputs (VAUXP[15:0] and VAUXN[15:0]) can be configured in this way. Setting a bit to logic 1 enables a bipolar input mode for the associated channel. Setting a bit to logic 0 (default) enables a unipolar input mode. All internal sensors use a unipolar transfer function.

ADC Channel Settling Time (4Eh and 4Fh)

The default settling time for an external channel in continuous sampling mode is four ADCCLK cycles. The settling time is additional acquisition time after the end of a conversion (see [Chapter 5, XADC Timing](#)). However, by setting the corresponding bits (for external channels) to logic 1 in registers 4Eh and 4Fh, the associated channel can have its settling time extended to 10 ADCCLK cycles. The bit definitions (which bits correspond to which external channels) for these registers are the same as the sequencer channel selection shown in [Table 4-1](#) and [Table 4-2](#).

Sequencer Modes

There are a number of sequencer modes as defined by [Table 3-9, page 38](#). These modes are described in this section.

Default Mode

The default mode is enabled by setting the sequence bits to 0. In this mode of operation, the XADC automatically monitors the on-chip sensors and stores the results in the status registers. Both ADCs are calibrated in this mode and an averaging of 16 samples is applied to all sensors. The XADC operates independently of any other control register settings in the mode. The XADC also operates in default mode during device configuration or if set using the sequence bits (see [Table 3-9](#)). [Table 4-3](#) shows the default sequence for the XADC. The automatic alarm function is not enabled in this mode.

Table 4-3: Default Mode Sequence

Order	Channel	Address	Description
1	Calibration	08h	Calibration of ADC A and ADC B
2	Temp	00h	Temperature sensor
3	V _{CCINT}	01h	V _{CCINT} supply sensor
4	V _{CCAUX}	02h	V _{CCAUX} supply sensor
5	V _{CCBRAM}	06h	V _{CCBRAM} supply sensor

Single Pass Mode

In single pass mode, the sequencer operates for one pass through the sequencer channel select registers (48h and 49h) and then halts. A sequence of channels as selected in these registers is converted. When the sequence bits as shown in [Table 3-9, page 38](#) are set so as to enable the automatic channel sequencer in single pass mode, the sequence starts. The

settings in sequencer registers 48h to 4Fh are used to operate the sequence in a user-defined mode of operation. All channels listed in [Table 4-1](#) and [Table 4-2](#) are available to be used in a sequence. For an explanation of the sequencer registers, see [Automatic Channel Sequencer](#). Another single pass can be started by writing to the sequence bits again. When the single pass is complete, the XADC defaults to Single Channel Mode described at the start of this chapter.

Continuous Sequence Mode

The continuous sequence mode is similar to single pass mode; however, the sequence automatically restarts as long as the mode is enabled.

The channel sequencer registers can also be reconfigured via the DRP at run time. The sequencer must first be disabled by writing to sequence bits SEQ3 to SEQ0 before writing to any of the sequencer channel registers. It is recommended that the XADC is placed in default mode by writing zeros to SEQ0 and SEQ1 while updating these registers. The XADC is automatically reset whenever SEQ3 to SEQ0 are changed. The current status register contents are not reset at this time. Restarting the sequencer by writing to bits SEQ3 to SEQ0 resets all channel averaging.

Simultaneous Sampling Mode

When placed in simultaneous sampling mode, the sequencer automatically sequences through eight pairs of auxiliary analog input channels for simultaneous sampling and conversion as shown in [Table 4-4](#). This is useful in applications where it is necessary to preserve the phase relationship between two signals.

Auxiliary analog channels 0 to 7 are assigned to ADC A and are nominated as *A channels*. Auxiliary analog channels 8 to 15 are assigned to ADC B and are nominated as *B channels* (see [Figure 1-1, page 10](#)). One A channel and one B channel are always sampled and converted at the same time in simultaneous sampling mode. [Table 4-4](#) shows how A and B pairs are selected using sequencer channel register 49h. Other sequencer registers that define averaging, analog input mode, and settling time are also available for use in this mode.

Channel Averaging (4Ah and 4Bh) and Analog-Input Modes (4Ch and 4Dh) can be set on a per channel basis in simultaneous sampling mode. Thus it is possible to have an A channel configured as unipolar and a B channel configured as bipolar. Bit definitions as shown in [Table 4-1](#) and [Table 4-2](#) apply. The settling time is applied to a channel pair only; thus setting bit 0 to 1 in register 4Eh sets the settling time to 10 ADCCLKs for both auxiliary channel 0 and channel 8.

The A and B channels are evenly spread across two I/O banks in the FPGA. Each bank typically supports 8 differential auxiliary analog input channels (using 16 I/Os). Thus each bank typically supports four A channels and four B channels.

Table 4-4: Sequencer Register (49h) Bit Definitions for Simultaneous Sampling Mode

Sequence Number	Bit	ADC Channel	Description
1	0	16, 24	Auxiliary channels 0 and 8
2	1	17, 25	Auxiliary channels 1 and 9
3	2	18, 26	Auxiliary channels 2 and 10
4	3	19, 27	Auxiliary channels 3 and 11

Table 4-4: Sequencer Register (49h) Bit Definitions for Simultaneous Sampling Mode (Cont'd)

Sequence Number	Bit	ADC Channel	Description
5	4	20, 28	Auxiliary channels 4 and 12
6	5	21, 29	Auxiliary channels 5 and 13 ⁽¹⁾
7	6	22, 30	Auxiliary channels 6 and 14 ⁽¹⁾
8	7	23, 31	Auxiliary channels 7 and 15 ⁽¹⁾
x	8	x	Undefined
x	9	x	Undefined
x	10	x	Undefined
x	11	x	Undefined
x	12	x	Undefined
x	13	x	Undefined
x	14	x	Undefined
x	15	x	Undefined

Notes:

1. These simultaneous sampling channels are not supported in Kintex-7 devices.

The on-chip temperature and power supplies can also be included in the sequence of channels monitored by setting the appropriate bits in Sequencer Register 48h. The second ADC (ADC B) does not carry out a conversion in simultaneous sampling mode when an internal sensor channel is being converted.

Automatic calibration cannot be enabled in simultaneous sampling mode. The XADC must be placed in default mode or other sequencer mode to carry out a calibration. The ADCs are automatically calibrated on power-up, which is sufficient for most applications.

Timing

The timing for simultaneous sampling mode is identical to other XADC modes (see [Chapter 5, XADC Timing](#)). Because the ADCs run in parallel, both status registers are updated at exactly the same time. Both continuous and event driven timing modes can be used.

Independent ADC Mode

In the independent ADC mode, ADC A is used to implement a fixed “monitoring mode” similar to the default mode except the alarm functions are enabled (see [Automatic Alarms, page 55](#)). In this mode, the alarm outputs are active, and the alarm threshold must be correctly configured by the user. Like default sequencer mode, the averaging is fixed at 16 samples.

ADC B is available to be used with the external analog input channels only. This mode frees up the second ADC for use in a customer application where only a single ADC is required while maintaining monitoring of the FPGA for reliability, safety, and anti-tamper applications.

Only the dedicated channel and auxiliary analog input channels can be assigned to ADC B in this sequencer mode. The internal channels (sensors) are automatically assigned to

ADC A, which automatically monitors these channels and generates alarms based on the user-defined alarm thresholds.

As with simultaneous sampling mode, it is not possible to select an automatic calibration of ADC B in this sequencer mode. ADC A is automatically calibrated in this mode. To carry out a calibration on ADC B and maintain on-chip monitoring, the default sequence mode should be selected. The XADC can then be returned to Independent ADC mode after EOS goes High at least once.

Sequencer Operation

Channel selection for independent ADC mode is defined using Sequencer Channel Registers 48h and 49h (see [Table 4-5](#) and [Table 4-6](#)). Internal sensors channels are automatically monitored and calibrated and cannot be assigned to the sequence for ADC B by users. Other sequencer registers that define settling time, analog input mode, and averaging remain unchanged.

Table 4-5: Independent ADC Sequencer Mode Bit Definitions (48h)

Sequence Number	Bit	ADC Channel	Description
-	0	-	Not Defined
	1		Not Defined
	2		Not Defined
	3		Not Defined
	4		Not Defined
	5		Not Defined
	6		Not Defined
	7		Not Defined
	8		Not Defined
	9		Not Defined
	10		Not Defined
1	11	3	V_P/V_N
--	12	-	Not Defined
	13		Not Defined
	14		Not Defined
	15		Not Defined

Table 4-6: Independent ADC Sequencer Mode Bit Definitions (49h)

Sequence Number	Bit	ADC Channel	Description
2	0	16	Auxiliary channel 0
3	1	17	Auxiliary channel 1
4	2	18	Auxiliary channel 2
5	3	19	Auxiliary channel 3
6	4	20	Auxiliary channel 4
7	5	21	Auxiliary channel 5
8	6	22	Auxiliary channel 6 ⁽¹⁾
9	7	23	Auxiliary channel 7 ⁽¹⁾
10	8	24	Auxiliary channel 8
11	9	25	Auxiliary channel 9
12	10	26	Auxiliary channel 10
13	11	27	Auxiliary channel 11
14	12	28	Auxiliary channel 12
15	13	29	Auxiliary channel 13 ⁽¹⁾
16	14	30	Auxiliary channel 14 ⁽¹⁾
17	15	31	Auxiliary channel 15 ⁽¹⁾

Notes:

1. Auxiliary channels 6, 7, 13, 14, and 15 are not supported on Kintex-7 devices.

Timing

The timing in this mode of operation is also the same for both ADCs. ADC A monitors the internal sensors at the same sample rate and mode selected for ADC B. Again, ADC B can be operated in continuous or event mode timing. In the case of event mode timing, internal sensor monitoring also stops if the CONVST signal is stopped. Therefore care should be taken to ensure periodic pulsing of CONVST even when ADC B is not being used by the application, if constant coverage of the on-chip environment is required.

Timing for continuous and event modes remain the same as shown in [Chapter 5, XADC Timing](#).

External Multiplexer Mode

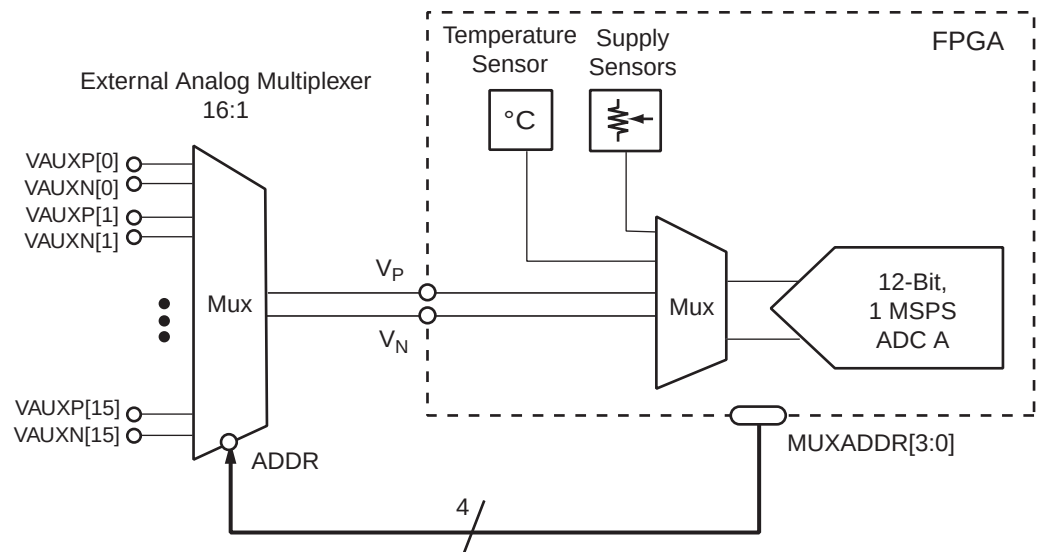
The XADC supports the use of an external analog multiplexer to implement a number of external analog inputs in situations where FPGA I/O resources are limited and auxiliary analog inputs are not available.

The XADC track/hold amplifiers return to track mode as soon as a conversion starts (see [Figure 4-3](#)). Therefore the acquisition on the next channel can start during the current conversion cycle. An output bus called MUXADDR[4:0] allows the XADC to control an external multiplexer. The address on this bus reflects the channel currently being acquired,

and it changes state as soon as the XADC enters acquisition mode. Users can also nominate the channel to be used with an external multiplexer.

External Multiplexer Operation

Figure 4-1 illustrates the external multiplexer concept. In this example an external 16:1 analog multiplexer is used instead of consuming the 32 FPGA I/Os required to implement the 16 auxiliary analog input channels using the internal multiplexer. Any four FPGA I/Os can be used for the external multiplexer decode operation. As shown in Figure 4-1, the dedicated analog inputs (V_P/V_N) are used to connect the external multiplexer to the XADC block, thereby making 16 analog inputs available. The external multiplexer mode of operation is enabled by setting the MUX bit in Configuration Register 0 (see [Control Registers](#), page 34).

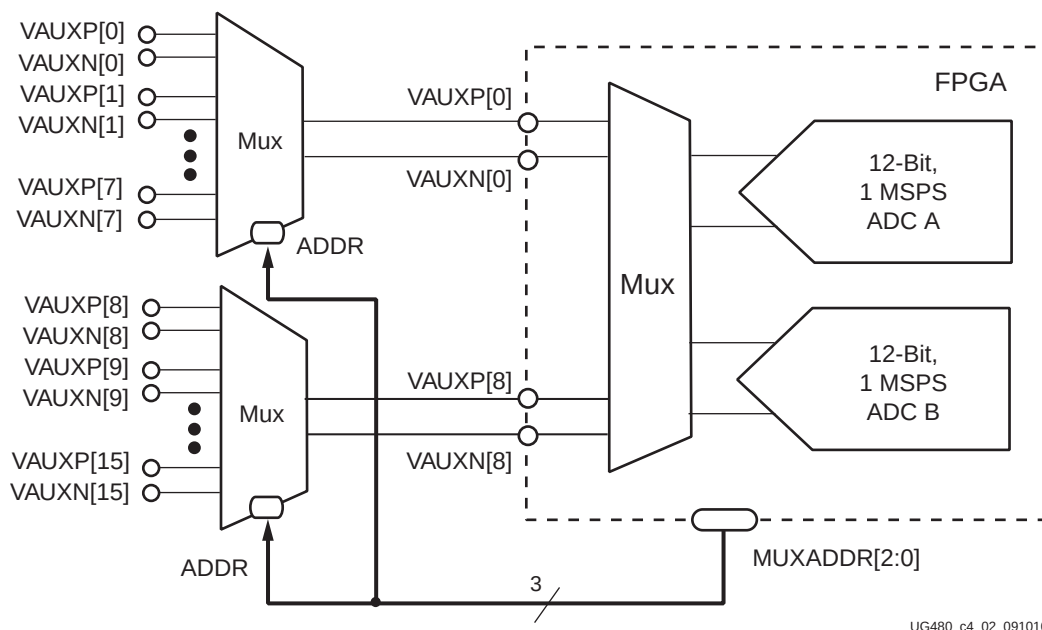


UG480_c4_01_091010

Figure 4-1: External Multiplexer Mode

When the MUX bit is set to a 1, the channel selection bits (CH0 to CH4) in Configuration Register 0 are used to nominate the channel for connection to the external multiplexer. For example, as shown in Figure 4-1, the dedicated analog input channel V_P/V_N is used. In this case, channel 3 (00011b) should be written to CH4 to CH0 in Control Register 40h. Any one of the auxiliary channels can also be used for connection to the external multiplexer. In the case of simultaneous sampling mode (see [Simultaneous Sampling Mode](#), page 49), two channels must be allocated to two external multiplexers to support simultaneous sampling.

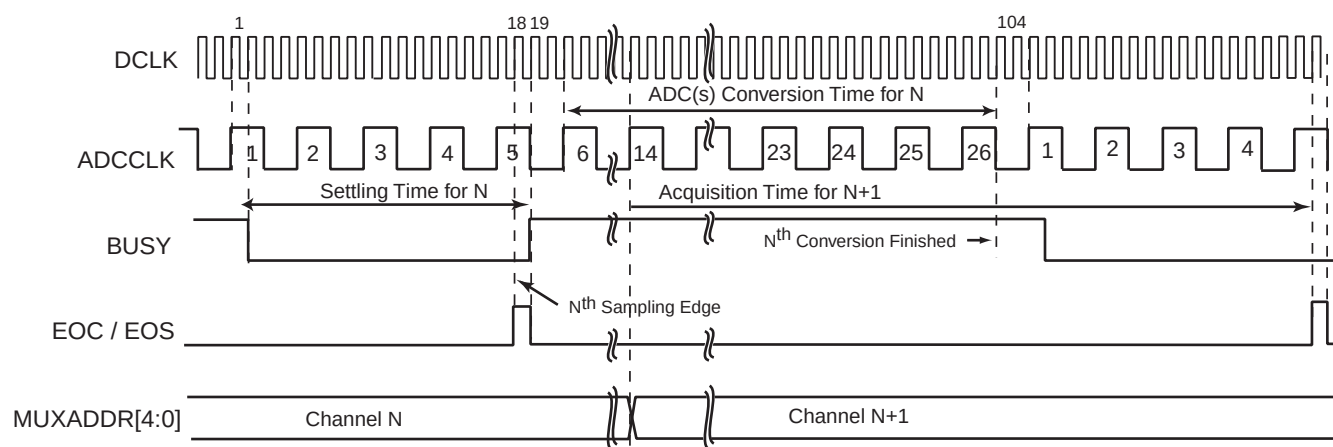
Figure 4-2 illustrates how the external multiplexer mode is implemented for simultaneous sampling mode. The channels selected for connection are also selected by writing to CH4 to CH0 but are allocated in pairs as defined in Table 4-4. For example, writing 16 (10000) to CH4 to CH0 would select auxiliary channels 0 and 8 for connection to external multiplexers as shown in Figure 4-2.



UG480_c4_02_091010

Figure 4-2: External Multiplexer Mode for Simultaneous Sampling

In both cases, the MUXADDR[4:0] bus is used to automatically select the external multiplexer channel. Figure 4-3 shows how the MUXADDR bus reflects the next channel selection ($N + 1$). MUXADDR changes state eight ADCCLK cycles after BUSY goes High.



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Figure 4-3: External Multiplexer Timing (Continuous Sampling Mode)

Maximum and Minimum Status Registers

The XADC also tracks the minimum and maximum values recorded for the internal sensors since the last power-up or since the last reset of the XADC control logic. The

maximum and minimum values recorded are stored in the DRP status registers starting at address 20h (see [Status Registers, page 32](#)). On power-up or after reset, all minimum registers are set to FFFFh, and all maximum registers are set to 0000h. Each new measurement generated for an on-chip sensor is compared to the contents of its maximum and minimum registers. If the measured value is greater than the contents of its maximum registers, the measured value is written to the maximum register. Similarly, for the minimum register, if the measured value is less than the contents of its minimum register, the measured value is written to the minimum register. This check is carried out every time a measurement result is written to the status registers.

Table 4-7: Maximum and Minimum Registers

Status Register	Description
20h	Temp Max - Maximum Temperature Recorded
21h	V _{CCINT} Max - Maximum V _{CCINT} Recorded
22h	V _{CCAUX} Max - Maximum V _{CCAUX} Recorded
23h	V _{CCBRAM} Max - Maximum V _{CCBRAM} Recorded
24h	Temp Min - Minimum Temperature Recorded
25h	V _{CCINT} Min - Minimum V _{CCINT} Recorded
26h	V _{CCAUX} Min - Minimum V _{CCAUX} Recorded
27h	V _{CCBRAM} Min - Minimum V _{CCBRAM} Recorded

Automatic Alarms

The XADC also generates an alarm signal on the logic outputs ALM[7:0] when an internal sensor measurement (Temperature, V_{CCINT}, V_{CCAUX}, or V_{CCBRAM}) exceeds some user-defined thresholds. Only the values written to the status registers are used to generate alarms. If averaging has been enabled for a sensor channel, the averaged value is compared to the Alarm Threshold register contents. The alarm outputs are disabled by writing a 1 to bits ALM3, ALM2, ALM1, and ALM0 in Configuration Register 1. The alarm thresholds are stored in Control Registers 50h to 5Fh. [Table 4-8](#) defines the alarm thresholds that are associated with specific control registers. The limits written to the threshold registers are MSB justified. Limits are derived from the temperature and power-supply sensor transfer functions (see [Figure 2-8, page 28](#) and [Figure 2-9, page 29](#)).

Table 4-8: Alarm Threshold Registers

Control Register	Description	Alarm
50h	Temperature Upper	ALM[0]
51h	V _{CCINT} Upper	ALM[1]
52h	V _{CCAUX} Upper	ALM[2]
53h	OT Upper	OT
54h	Temperature Lower	ALM[0]
55h	V _{CCINT} Lower	ALM[1]
56h	V _{CCAUX} Lower	ALM[2]

Table 4-8: Alarm Threshold Registers (Cont'd)

Control Register	Description	Alarm
57h	OT Lower	OT
58h	V _{CCBRAM} Upper	ALM[3]
59h	Not Defined	-
5Ah	Not Defined	-
5Bh	Not Defined	-
5Ch	V _{CCBRAM} Lower	ALM[3]
5Dh	Not Defined	-
5Eh	Not Defined	-
5Fh	OT Lower	-

Supply Sensor Alarms

When the measured value on the supply sensor for V_{CCINT}, V_{CCAUX}, or V_{CCBRAM} is greater than the thresholds in Control Registers 51h, 52h, and 58h or less than the thresholds in Control Registers 55h, 56h, and 5Ch, respectively, the output alarms go active. The alarms are reset when a subsequently measured value falls inside the threshold.

Thermal Management

The on-chip temperature measurement is used for critical temperature warnings. The default Over Temperature threshold is 125°C. This threshold is used for preconfiguration or when the contents of the OT Upper alarm register have not been configured (see Table 4-8). To override this default condition, the 12 MSBs of the OT Upper Register (Control Register 53h) must be set using the temperature sensor transfer function (see Figure 2-8). In addition, the four LSBs must be set to 0011b.

When the die temperature exceeds a factory set limit of 125°C or a user-defined threshold, the Over-Temperature alarm logic output (OT) becomes active. This feature can be disabled by the user when the XADC is instantiated in a design. The OT function is disabled by writing a 1 to the OT bit in Configuration Register 1. The OT signal resets when the FPGA temperature has fallen below a user-programmable limit in Control Register 57h (see Table 4-8). When the automatic power-down feature is enabled, the OT signal can be used to trigger a device power down. When OT goes High, the FPGA enters power down approximately 10 ms later. The power-down feature initiates a configuration shutdown sequence, disabling the device when finished and asserts GHIGH to prevent any contention (see the *7 Series FPGAs Configuration User Guide*). When OT is deasserted, GHIGH also is deasserted and the start-up sequence is initiated releasing all global resources. By default this functionality is disabled and must be explicitly enabled.

The automatic power down is enabled by using a configuration option in the ISE® software. To generate a programming file, the user checks the “Power Down Device if Over Safe Temperature” option under “Configuration Options” on the Process Properties GUI. Alternatively, the user can use the **bitgen -g OverTempPowerDown: [Enable|Disable]** command line option. When the FPGA enters power down, the XADC continues to operate in whatever mode was configured prior to power down using an internal clock oscillator. The FPGA automatically powers up once the temperature has fallen below the user-programmable lower limit. The XADC OT signal can also be reset by

writing a 1 to the OT bit in XADC Configuration Register 1 via the JTAG DRP interface. On-chip sensors are monitored via the JTAG TAP during device power down. During power down, the XADC uses an internal oscillator instead of DCLK. On power-up, several DCLK cycles are required to resynchronize the DRP (JTAGLOCKED remains High until the DRP is ready for use again).

A second user-programmable temperature threshold level (Control Register 50h) is used to carry out a user-defined thermal management procedure, such as powering on or controlling the speed of a fan. An alarm signal (ALM[0]) becomes active when the FPGA temperature exceeds the limit in this register. The signal is available through the interconnect and is routed using the FPGA resources. The alarm signal resets when the temperature falls below the threshold in Control Register 54h. This operation differs for the supply sensor alarm, because the supply alarm resets when the measurement is between the upper and lower thresholds.

XADC Timing

All XADC timing is synchronized to the DRP clock (DCLK). The ADCCLK is generated by dividing DCLK by the user selection in Configuration Register 2 (see [Control Registers, page 34](#)). ADCCLK is an internal clock used by the ADCs and is not available externally. ADCCLK is only included here to aid in describing the timing.

The ADC block is operated in one of two possible timing modes, continuous sampling mode and event driven sampling mode.

In continuous sampling mode, the ADC automatically starts a new conversion at the end of a current conversion cycle. In event sampling mode, the user must initiate the next conversion after the current conversion cycle ends by using the CONVST or CONVSTCLK inputs. The operating mode is selected by writing to Configuration Register 0 (see [Control Registers](#)).

Refer to the respective 7 series FPGAs data sheet for the latest XADC timing specifications. The robust nature of the XADC ensures continued and correct operation even if the external clock input DCLK is stopped. In this situation, the XADC automatically switches over to an internal clock oscillator to continue to operate as configured. After configuration, the DCLK input requires 20 DCLKs to resynchronize to the external clock. The JTAGLOCKED signal is High during this period. It is not possible to access the DRP until resynchronization has occurred.

Continuous Sampling

In continuous sampling mode, the ADCs continue to carry out a conversion on the selected analog inputs. [Figure 5-1](#) shows the timing associated with continuous sampling mode. The ADCCLK is generated by a clock divider. The analog-to-digital conversion process is made up of two parts, the acquisition phase and the conversion phase.

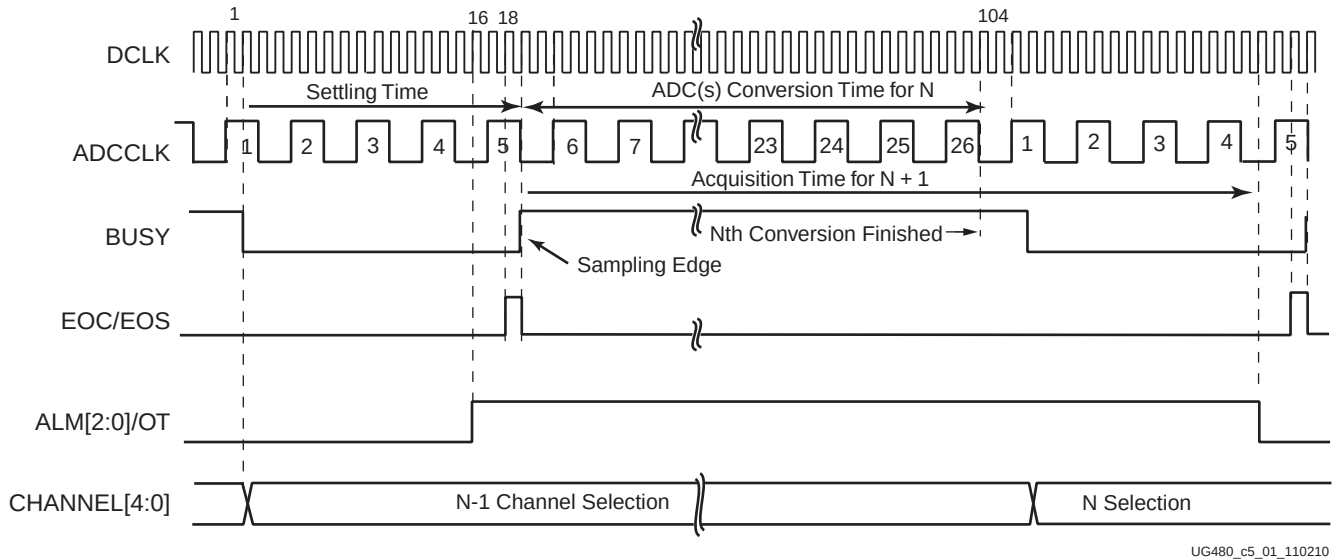


Figure 5-1: Continuous Sampling Mode

Acquisition Phase

During the acquisition phase, the ADC acquires the voltage on a selected channel to perform the conversion. The acquisition phase involves charging a capacitor in the ADC to the voltage on the selected channel (see [Analog Input Description, page 25](#) for more information). The time required to charge this capacitor depends on the selected input-channel source impedance. The XADC allows the acquisition of the next channel to start during the current conversion. This is possible because the ADC has a separate Track-and-Hold amplifier (T/H). Thus when the ADC starts to convert an input voltage, the T/H is free to start precharging to the next voltage to be converted. This allows for a faster sampling rate because the ADC does not need to acquire the full input voltage at the end of the current conversion. By default a *settling period* of four ADCCLK cycles is left between the end of the current conversion and the start of the next conversion. This can be extended to 10 ADCCLK cycles by setting the ACQ bits in either the control registers or the sequence registers.

When operating in single channel mode (see [Chapter 4, XADC Operating Modes](#)), the user must write to Configuration Register 0 to select the next channel for conversion. The address of the next channel to convert is read from the control registers at the end of the previous conversion (when BUSY goes Low), and the analog signal is acquired during the current conversion. Additional acquisition or settling time after the end of the current conversion is configured by the ACQ bit in Configuration Register 0.

For more information on the effects of source impedance on the acquisition, see [Analog Input Description](#).

Conversion Phase

The conversion phase starts on the sampling edge (next rising edge of DCLK) at the end of the 4 or 10 ADCCLK cycles settling time. The BUSY signal transitions to an active High on the next rising edge of DCLK to indicate the ADC is carrying out a conversion. The conversion phase is 22 ADCCLK cycles long. The conversion result transfers to the status registers 16 DCLK cycles after BUSY transitions Low, and the EOC logic output pulses High for one DCLK cycle (see [Figure 5-1](#)). If the channel being converted is also being

averaged, then the averaged data is only transferred to the status registers when the last sample result has been converted. Thus, if a channel is being averaged, an EOC pulse is only generated for the last conversion result, such as the 16th, 64th, 256th sample. When averaging is enabled, write operations to the configuration registers are not acted on until EOC is pulsed, i.e., when the averaging operation is complete.

When XADC is being operated in a sequence mode, the user identifies the channel being converted by monitoring the channel address (CHANNEL[4:0]) logic outputs. The multiplexer channel address of the channel being converted is updated on these logic outputs when BUSY transitions Low at the end of the conversion phase. The channel address outputs can be used with the EOC and DRDY signals to automatically latch the contents of the output data registers into a FIFO or block RAM. This is accomplished by connecting the CHANNEL[4:0] outputs to DADDR[4:0] (with DADDR[6:5] = 0), using EOC as a DEN (enable) for the DRP, and using DRDY as a WE (write enable) for the block RAM.

XADC EOS signal has the same timing as EOC. This signal is pulsed when the output data register for the last channel in a programmed channel sequence is updated.

Event-Driven Sampling

Figure 5-2 illustrates the event-driven sampling mode. In this operating mode, the sampling instant and subsequent conversion process are initiated by a trigger signal called convert start (CONVST). Event-driven sampling mode is used when precise control over the sampling instant is necessary.

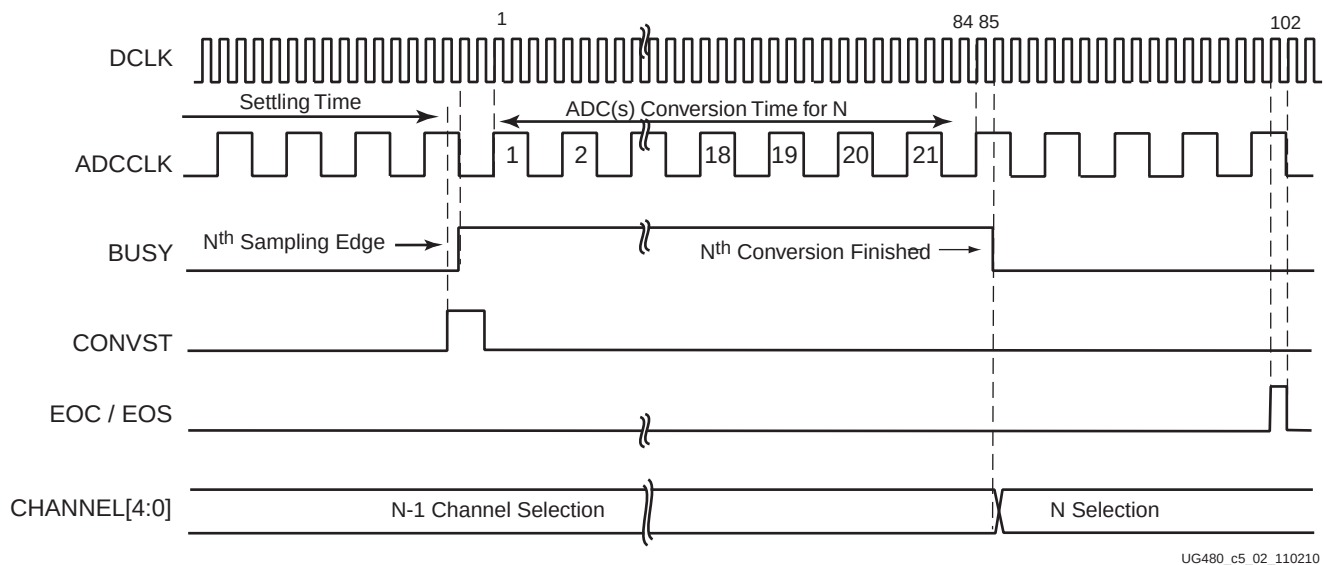


Figure 5-2: Event Driven Sampling Mode

The DCLK must always be present when using event-driven sampling mode. If no DCLK is present, the XADC reverts to continuous mode timing using an internal clock oscillator.

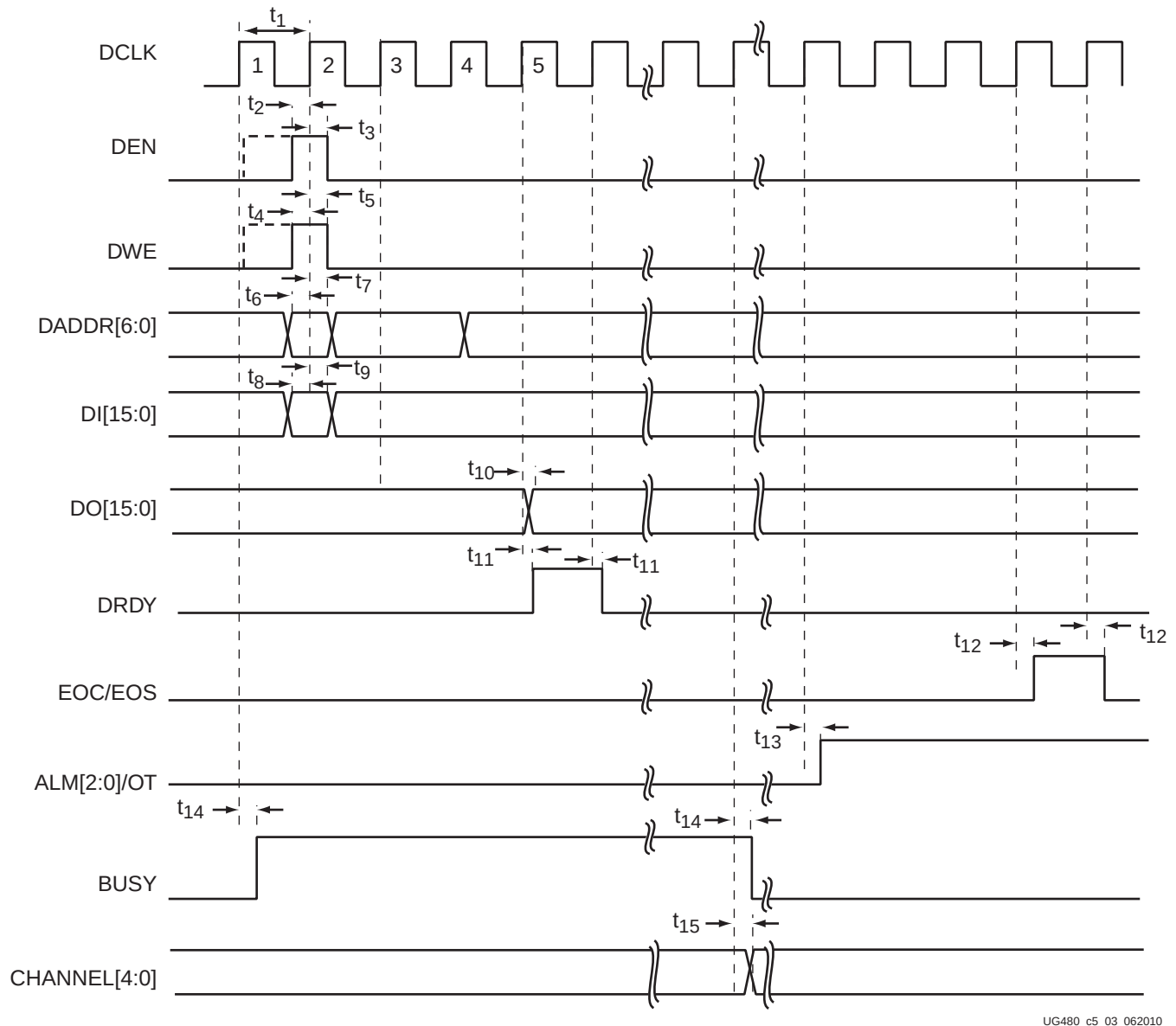
A Low-to-High transition (rising edge) on CONVST or CONVSTCLK defines the exact sampling instant for the selected analog-input channel. The BUSY signal transitions High just after the sampling instant on the next rising edge of DCLK. CONVST can be an asynchronous external signal in which case the XADC automatically resynchronizes the conversion process to the ADCCLK.

As with the continuous sampling mode, enough time must be allowed for the acquisition phase, that is, the time between a channel change and the sampling edge (the rising edge of CONVST or CONVSTCLK, see [Analog Input Description, page 25](#)). This allows the ADC to acquire the new signal before it is sampled by the CONVST signal and the conversion phase starts. The ACQ bit has no meaning in event-sampling mode because the sampling instant is controlled by CONVST/CONVSTCLK. Therefore, the acquisition time on a selected channel is also controlled by CONVST/CONVSTCLK. CONVST and CONVSTCLK are logically ORed within the XADC. The T/H starts to acquire the voltage on the next channel as soon as BUSY goes High and a conversion starts.

After the analog input has been sampled by a rising edge on CONVST/CONVSTCLK, a conversion is initiated on the next rising edge of ADCCLK. After a conversion has been initiated by CONVST, it is not possible to interrupt the conversion or start a new conversion until BUSY goes Low.

After BUSY goes Low, the conversion result is transferred to the output status registers 16 DCLK cycles later, and the EOC logic output pulses High for one DCLK cycle at this time. If the channel being converted is also being filtered, the averaged data is only transferred to the status registers when the last sample result has been converted. Thus, if a channel is being averaged, no EOC pulse is generated for all but the last conversion result (such as the 16th, 64th, and 256th sample), depending on the filter setting. The EOC, EOS, and CHANNEL[4:0] outputs operate in the same way as in the continuous sampling mode, (see [Continuous Sampling](#)).

Table 5-1 describes the XADC timing event shown in Figure 5-3.



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Figure 5-3: XADC Detailed Timing

Table 5-1: XADC Timing Information

Event	Description
t_1	DCLK period ⁽¹⁾
t_2	Minimum DEN setup time before the rising edge of DCLK ⁽²⁾
t_3	Minimum DEN hold time after the falling edge of DCLK
t_4	Minimum DWE setup time before the rising edge of DCLK ⁽³⁾
t_5	Minimum DWE hold time after the falling edge of DCLK ⁽³⁾

Table 5-1: XADC Timing Information (Cont'd)

Event	Description
t_6	Minimum DRP address setup time before the rising edge of DCLK
t_7	Minimum DRP address hold time after the rising edge of DCLK
t_8	Minimum DRP input data bus setup time before the rising edge of DCLK
t_9	Minimum DRP input data bus hold time after the rising edge of DCLK
t_{10}	Maximum DRP DCLK to DO delay/access time ⁽⁴⁾
t_{11}	Maximum delay on DRDY going High/Low after a rising edge on DCLK
t_{12}	Maximum delay on EOC/EOS going High/Low after a rising edge on DCLK
t_{13}	Maximum delay on ALM[2:0]/OT going High after a rising edge on DCLK
t_{14}	Maximum delay on BUSY going High/Low after a rising edge on DCLK
t_{15}	Maximum delay on CHANNEL[4:0] on the rising edge of DCLK

Notes:

1. The minimum DCLK duty cycle is 60/40.
2. DEN should only be pulsed for one DCLK cycle.
3. For a DRP write operation, the DADDR[6:0] address and DI[15:0] are latched on the rising edge of DCLK when DEN and WEN are High. The data is placed in the DRP register three DCLK cycles later. DRDY goes High when the data has been written. See the Dynamic Reconfiguration Port description in the *7 Series FPGAs Configuration User Guide* for more information.
4. For a DRP read operation, the DADDR[6:0] address is latched on the rising edge of DCLK when DEN is High. Three DCLK cycles later, the data is placed on the DO bus. DRDY goes High when the data on DO is valid. See the Dynamic Reconfiguration Port description in the *7 Series FPGAs Configuration User Guide* for more information.

Application Guidelines

The 7 series FPGAs XADC is a precision analog measurement system based on a 12-bit analog-to-digital converter (ADC) with an LSB size approximately equal to 250 μ V. To achieve the best possible performance and accuracy with all measurements (both on-chip and external), a number of dedicated pins for the ADC reference and power supply are provided. Care must be taken with the connection of these pins to ensure the best possible performance from the ADC. This chapter outlines the basic design guidelines to consider as part of the requirements for board design.

Reference Inputs (V_{REFP} and V_{REFN})

These high-impedance inputs are used to deliver a differential reference voltage for the analog-to-digital conversion process. ADCs are only as accurate as the reference provided. Any reference-voltage error results in a gain error vs. the ideal ADC transfer function (see [Chapter 2, Analog-to-Digital Converter](#)). Errors in the reference voltage affect the accuracy of absolute measurements for both on-chip sensors and external channels.

For typical usage, the reference voltage between V_{REFP} and V_{REFN} should be maintained at 1.25V $\pm$ 0.2% using an external reference IC. Reference voltage ICs that deliver 1.25V are widely available from several vendors. Many vendors offer reference voltage ICs in small packages (SOT-23 and SC70). The 1.25V reference should be placed as close as possible and connected directly to the V_{REFP} input, using the decoupling capacitors recommended in the reference IC data sheet. The recommended reference connections are illustrated in [Figure 6-1](#).

The XADC also has an on-chip reference option which is selected by connecting V_{REFP} and V_{REFN} to AGND as shown in [Figure 6-1](#). Due to reduced accuracy, the on-chip reference does impact the measurement performance of the XADC as explained in this section. The performance with on-chip reference is specified in the respective 7 series FPGAs data sheet.

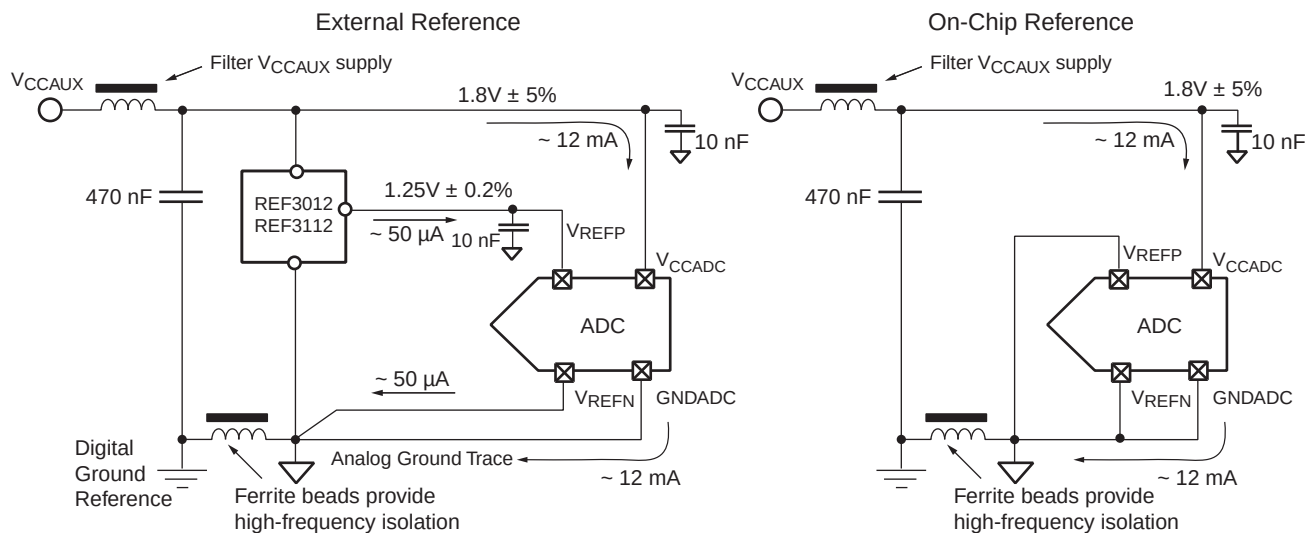
Analog Power Supply and Ground (V_{CCADC} and GNDADC)

These inputs provide the power supply and ground reference for the analog circuitry in the XADC. A common mechanism for the coupling of noise into an analog circuit is from the power supply and ground connections. Excessive noise on the analog supply or ground reference affects the ADC measurement accuracy. For example, I/O switching activity can cause significant disturbance of the digital ground reference plane. Thus, it would not be advisable to use the digital ground as an analog ground reference for XADC.

Similarly, for the digital supplies for the FPGA logic, high switching rates easily result in high-frequency voltage variations on the supply, even with decoupling. In an effort to mitigate these effects on the ADC performance, a dedicated supply and ground reference

is provided. Figure 6-1 illustrates how to use the 1.8V V_{CCAUX} supply to power the analog circuitry. V_{CCAUX} is filtered using a low-pass network. The filter design depends on the ripple and ripple frequency (if any) on the V_{CCAUX} supply if, for example, a switching regulator is used. There is also a power-supply rejection specification for the external reference circuit which needs to be considered. The filtering should ensure no more than 1 LSB (250 μ V) of noise on the reference output to minimize any impact on ADC accuracy at 12 bits.

The other source of noise coupling into the ADC is from the ground reference GNDADC. In mixed-signal designs, it is common practice to use a separate analog ground plane for analog circuits to isolate the analog and digital ground return paths to the supply. Common ground impedance is a mechanism for noise coupling and needs to be carefully considered when designing the PCB. This is shown in Figure 2-3, page 24, where the common ground impedance R_G converts digital switching currents into a noise voltage for the analog circuitry. While a separate analog ground plane is recommended for 12-bit operation, it is often not possible or practical to implement a separate analog ground plane in a design. For example, if a user only intends to use the on-chip sensors, one low-cost solution is to isolate V_{REFN} and GNDADC ground references (such as a trace) from the digital ground (plane) using a ferrite bead as shown in Figure 6-1. This is illustrated in PC Board Design Guidelines.



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Figure 6-1: ADC Power and Ground Connections

The ferrite bead behaves like a resistor at high frequencies and functions as a lossy inductor. A typical ferrite impedance vs. frequency plot is shown in Figure 6-2. The ferrite helps provide high frequency isolation between digital and analog grounds. The reference IC maintains a 1.25V difference of between V_{REFP} and V_{REFN} . The ferrite offers little resistance to the analog DC return current.

The reference inputs should be routed as a tightly coupled differential pair from the reference IC to the package pins. If routed on the same signal layer, the supply and analog ground traces (V_{CCADC} and GNDADC) should be used to shield the reference inputs because they have a higher tolerance to any coupled noise.

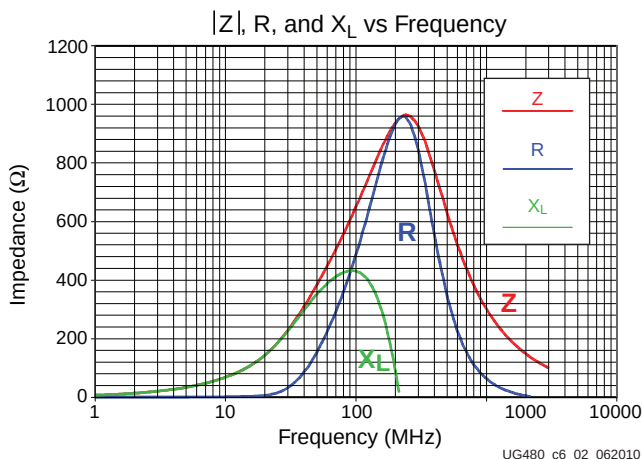


Figure 6-2: Ferrite Bead Characteristic

External Analog Inputs

The analog inputs are high-impedance differential inputs. The differential input scheme enables the rejection on common mode noise on any externally applied analog-input signal. Because of the high impedance of each input (such as V_P and V_N), the input AC impedance is typically determined by the sensor, the output impedance of the driving circuitry, or other external components. Figure 6-3 illustrates a simple resistor divider network is used to monitor an external 2.5V supply rail in unipolar input mode. To ensure that noise coupled onto the analog inputs is common to both inputs (reduce differential noise), the impedance on each input should be matched. Analog-input traces on the PCB should also be routed as tightly coupled differential pairs.

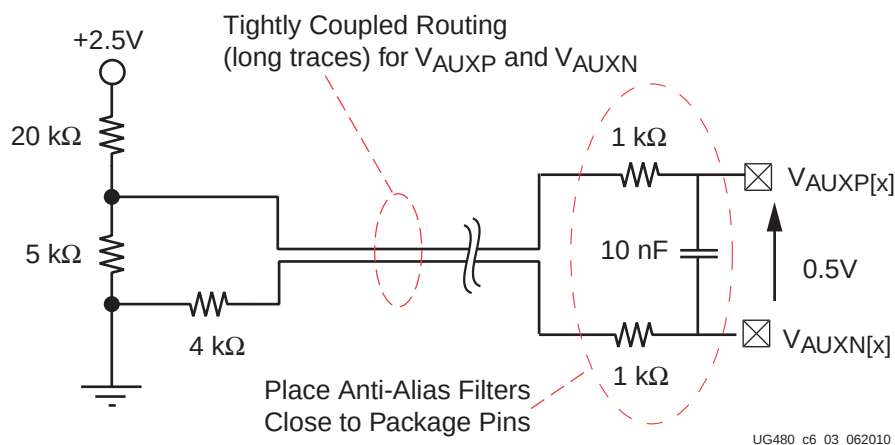


Figure 6-3: Voltage Attenuation

Anti-Alias Filters

Also shown in Figure 6-3, is a low-pass filter network at the analog differential inputs. This filter network is commonly referred to as the anti-alias filter and should be placed as close as possible to the package pins. The sensor can be placed remotely from the package as long as the differential input traces are closely coupled. The anti-alias filter attenuates

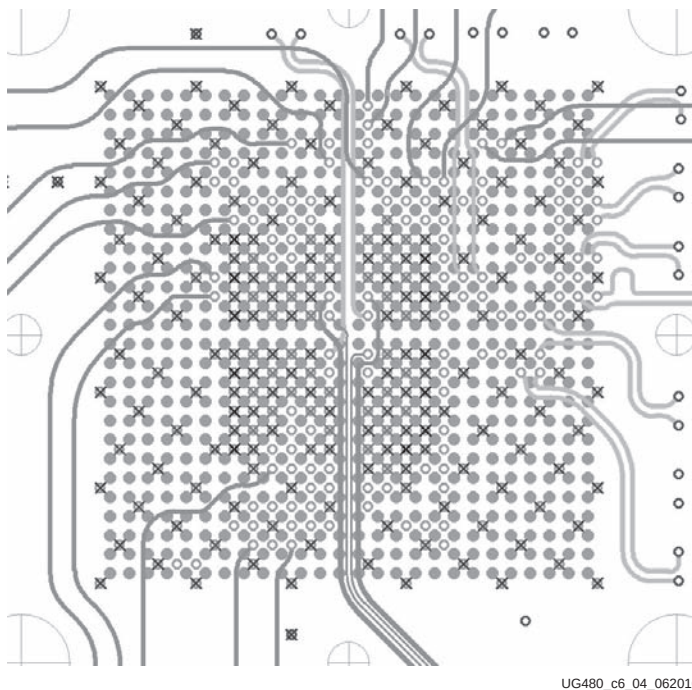
high-frequency signal components entering the ADC where they could be sampled and aliased, resulting in ADC measurement corruption. A discussion of aliasing in sampled systems is beyond the scope of this document. A good data converter reference book can provide more information on this topic.

Over and Under Voltages

The input voltage can exceed VCCADC (1.8V) or go below GNDADC by as much as 100 mV without damage to the XADC. A current-limiting resistor of at least 100 Ω should be placed in series with the analog inputs. The resistors in the anti-alias filters fulfill this requirement. If the analog input range (1V) is exceeded, the ADC output code clips at the maximum output code shown in [Figure 2-1, page 22](#) or [Figure 2-2, page 23](#), depending on the analog input mode. Negative input voltages clip at zero code.

PC Board Design Guidelines

[Figure 6-4](#) and [Figure 6-5](#) illustrate one possible way to address the requirements outlined in the previous sections. [Figure 6-4](#) shows how, by staggering the vias with respect to the pads, north-south and east-west routing channels through the via field are created. These routing channels can be used to bring tightly coupled differential pairs into the center of the via field—even when using 5 mil tolerances.



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Figure 6-4: Routing Channels to Center of Array Created by Staggering Vias

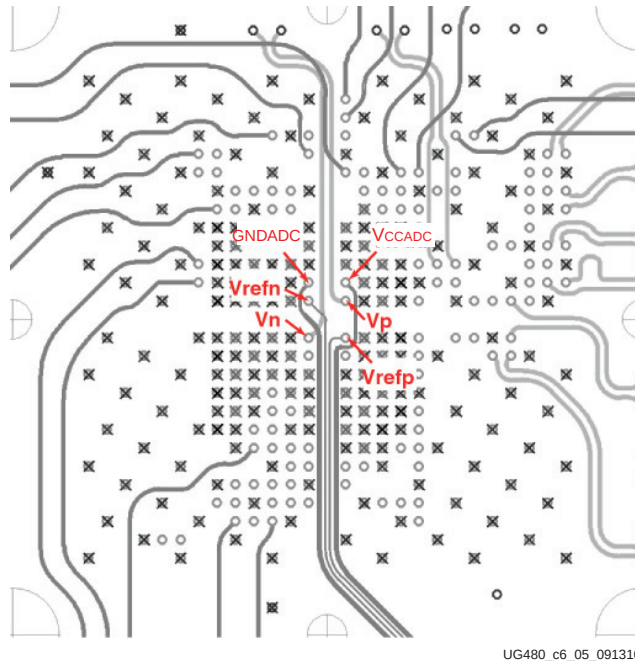


Figure 6-5: Reference Inputs (V_{REFP} and V_{REFN}) should be Routed as Differential Pairs into the Center of the BGA

In Figure 6-5, the pads have been removed for clarity. The reference inputs (V_{REFP} and V_{REFN}) are routed as a tightly coupled differential pair from an external 1.25V reference IC at the bottom edge of the FPGA (refer to Figure 6-1 for the connections). The analog power supply and ground reference are also routed into the center of the BGA using traces. No power planes are required to supply a ground reference for the XADC. The analog supply and ground reference are connected to the external reference IC as shown in Figure 6-1. In the PCB implementation shown, the supply and ground traces are routed on either side of the reference traces on the same signal layer and act as guards between the reference traces and any potential aggressors (e.g., clocks and switching I/Os). It is not a requirement that the supply and ground traces are routed on the same signal layer as shown, but they should be routed on an adjacent layer. The V_{REFN} and GNDADC traces should be connected at (or close to) the ground pin of the reference IC. The ferrite bead that connects the analog ground trace to system ground should also be placed close to the reference IC. Figure 6-5 also shows the routing (from the top) of the dedicated analog input pair (V_P and V_N). These inputs are also routed as a differential pair.

The external reference IC should be placed as close as possible to the FPGA to reduce the opportunities for coupled noise and to minimize any impedances in the reference traces. The staggered via field also allows the 10 nF decoupling on V_{REFP} and V_{CCADC} to be placed in the center of the array close to the package balls. V_{REFP} should be decoupled to V_{REFN} and V_{CCADC} to GNDADC near the package balls.

When using the on-chip reference, the layout of the PCB is greatly simplified. The V_{REFP} and V_{REFN} pins should be shorted to AGND locally at the package balls. The ferrite beads used to separate AGND and digital GND should be placed close to the XADC balls in the center of the array along with a 10 nF decoupling capacitor for V_{CCADC} .

These suggestions are only intended to guide a low-cost PC board implementation. If it is feasible to create an analog reference plane, then there is no issue with doing this and it helps improve the ADC performance. However, the reference inputs should still be routed as differential pairs as shown.

XADC Software Support

This section explains in more detail the example shown in [Example Instantiation in Chapter 1](#). The Verilog source code and test bench for this example can be found in the ZIP (ug480_7Series_XADC.zip) file which accompanies this user guide. In addition to instantiating a basic design, the use of the Analog Stimulus File as a means to introduce analog signals into the simulation is also explained. Finally, the output of a simple behavioral simulation of the design is shown.

Example Design

In this example, the XADC is set up to monitor all on-chip sensors (i.e., Temperature, V_{CCINT} , V_{CCAUX} , and V_{CCBRAM}). See [Temperature Sensor, page 18](#) and [Power Supply Sensors, page 19](#). The XADC is also set to automatically generate alarm outputs when the defined operating ranges for the FPGA are exceeded (see [Automatic Alarms, page 55](#)). The XADC is also operated in Continuous Sequence mode for this example (see [Sequencer Modes, page 48](#)). For clarity (and shorter simulations), the averaging function has not been enabled in this example. Averaging does not have any impact on the simulation results because an ideal model of the XADC is used; however, it is recommended that averaging of at least 16 samples is used in practical designs.

Apart from initializing the Alarm Threshold Registers and the Automatic Channel Sequencer Register, the Configuration Registers need to be initialized to enable alarm outputs, sequencer modes, and ADC clock dividers (see [Configuration Registers \(40h to 42h\), page 35](#) for more information). Here is an instantiation in Verilog of the example design:

```
`timescale 1ns / 1ps
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
// Company:      Xilinx Inc.
//
// Create Date:   09/02/2010
//
// Module Name:   My_XADC
// Project Name:  XADC UG Example
// Target Devices: Virtex-7
// Tool versions: 13.0.2 Beta 2
// Description:   Example XADC instantiation monitoring on chip sensors and
//               auxiliary analog input channels 0 - 3
// Dependencies:
//
// Revision:      1.0
// Comments:      See 7 series XADC User Guide
//
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////

module My_XADC
(
  DADDR_IN,          // Address bus for the dynamic reconfiguration port
  DCLK_IN,           // Clock input for the dynamic reconfiguration port
  DEN_IN,            // Enable Signal for the dynamic reconfiguration port
  DI_IN,             // Input data bus for the dynamic reconfiguration port
  DWE_IN,            // Write Enable for the dynamic reconfiguration port
  RESET_IN,          // Reset signal for the System Monitor control logic
  VAUXP0,            // Auxiliary channel 0
  VAUXN0,
  VAUXP1,            // Auxiliary channel 1
  VAUXN1,
  VAUXP2,            // Auxiliary channel 2
  VAUXN2,
```



```

VAUXP3,          // Auxiliary channel 3
VAUXN3,
BUSY_OUT          // ADC Busy signal
CHANNEL_OUT,      // Channel Selection Outputs
DO_OUT,           // Output data bus for dynamic reconfiguration port
DRDY_OUT,         // Data ready signal for the dynamic reconfiguration port
EOC_OUT,          // End of Conversion Signal
EOS_OUT,          // End of Sequence Signal
OT_OUT,           // Over-Temperature alarm output
VCCAUX_ALARM_OUT, // VCCAUX-sensor alarm output
VCCINT_ALARM_OUT, // VCCINT-sensor alarm output
USER_TEMP_ALARM_OUT, // Temperature-sensor alarm output
VBRAM_ALARM_OUT,  // VCCBRAM-sensor alarm output
ALARM_OUT,        // OR'ed output of all the Alarms
VP_IN,            // Dedicated Analog Input Pair
VN_IN);

input [6:0] DADDR_IN;
input DCLK_IN;
input DEN_IN;
input [15:0] DI_IN;
input DWE_IN;
input RESET_IN;
input VAUXP0;
input VAUXN0;
input VAUXP1;
input VAUXN1;
input VAUXP2;
input VAUXN2;
input VAUXP3;
input VAUXN3;
input VP_IN;
input VN_IN;

output BUSY_OUT;
output [4:0] CHANNEL_OUT;
output [15:0] DO_OUT;
output DRDY_OUT;
output EOC_OUT;
output EOS_OUT;
output OT_OUT;
output VCCAUX_ALARM_OUT;
output VCCINT_ALARM_OUT;
output USER_TEMP_ALARM_OUT;
output VBRAM_ALARM_OUT;
output ALARM_OUT;

wire GND_BIT;
assign GND_BIT = 0;
wire [15:0] aux_channel_p;
wire [15:0] aux_channel_n;
wire [7:0] alm_int;

assign ALARM_OUT = alm_int[7];
assign VBRAM_ALARM_OUT = alm_int[3];
assign VCCAUX_ALARM_OUT = alm_int[2];
assign VCCINT_ALARM_OUT = alm_int[1];
assign USER_TEMP_ALARM_OUT = alm_int[0];

assign aux_channel_p[0] = VAUXP0;
assign aux_channel_n[0] = VAUXN0;

assign aux_channel_p[1] = VAUXP1;
assign aux_channel_n[1] = VAUXN1;

```

```

assign aux_channel_p[2] = VAUXP2;
assign aux_channel_n[2] = VAUXN2;

assign aux_channel_p[3] = VAUXP3;
assign aux_channel_n[3] = VAUXN3;

assign aux_channel_p[4] = 1'b0;
assign aux_channel_n[4] = 1'b0;

assign aux_channel_p[5] = 1'b0;
assign aux_channel_n[5] = 1'b0;

assign aux_channel_p[6] = 1'b0;
assign aux_channel_n[6] = 1'b0;

assign aux_channel_p[7] = 1'b0;
assign aux_channel_n[7] = 1'b0;

assign aux_channel_p[8] = 1'b0;
assign aux_channel_n[8] = 1'b0;

assign aux_channel_p[9] = 1'b0;
assign aux_channel_n[9] = 1'b0;

assign aux_channel_p[10] = 1'b0;
assign aux_channel_n[10] = 1'b0;

assign aux_channel_p[11] = 1'b0;
assign aux_channel_n[11] = 1'b0;

assign aux_channel_p[12] = 1'b0;
assign aux_channel_n[12] = 1'b0;

assign aux_channel_p[13] = 1'b0;
assign aux_channel_n[13] = 1'b0;

assign aux_channel_p[14] = 1'b0;
assign aux_channel_n[14] = 1'b0;

assign aux_channel_p[15] = 1'b0;
assign aux_channel_n[15] = 1'b0;

XADC #(
    // Initializing the XADC Control Registers
    .INIT_40(16'h0000), // config reg 0
    .INIT_41(16'h2ef0), // Continuous Sequencer Mode, Disable unused ALMs,
                        // and Enable calibration
    .INIT_42(16'h0400), // Set DCLK divider to 4, ADC = 500Ksps, DCLK = 50MHz
    .INIT_48(16'h4701), // Sequencer channel - enable on-chip sensors and calibration
    .INIT_49(16'h000f), // Sequencer channel - enable aux channels 0 - 3
    .INIT_4A(16'h0000), // Sequencer Average selection
    .INIT_4B(16'h0000), // Sequencer Average selection
    .INIT_4C(16'h0000), // Sequencer Bipolar selection
    .INIT_4D(16'h0000), // Sequencer Bipolar selection
    .INIT_4E(16'h0000), // Sequencer Acq time selection
    .INIT_4F(16'h0000), // Sequencer Acq time selection
    .INIT_50(16'hb5ed), // Temp upper alarm trigger 85°C
    .INIT_51(16'h5999), // Vccint upper alarm limit 1.05V
    .INIT_52(16'ha147), // Vccaux upper alarm limit 1.89V
    .INIT_53(16'hdddd), // OT upper alarm limit 125°C
    .INIT_54(16'ha93a), // Temp lower alarm reset 60°C
    .INIT_55(16'h5111), // Vccint lower alarm limit 0.95V
    .INIT_56(16'h91Eb), // Vccaux lower alarm limit 1.71V
    .INIT_57(16'hae4e), // OT lower alarm reset 70°C
    .INIT_58(16'h5999), // VBRAM upper alarm limit 1.05V

```



```

.INIT_5C(16'h5111),      // VBRAM lower alarm limit 0.95V
.SIM_MONITOR_FILE("sensor_input.txt") // Analog Stimulus file
)

XADC_INST (// See UG for port descriptions
.CONVST(GND_BIT),
.CONVSTCLK(GND_BIT),
.DADDR(DADDR_IN[6:0]),
.DCLK(DCLK_IN),
.DEN(DEN_IN),
.DI(DI_IN[15:0]),
.DWE(DWE_IN),
.RESET(RESET_IN),
.VAUXN(aux_channel_n[15:0]),
.VAUXP(aux_channel_p[15:0]),
.ALM(alm_int),
.BUSY(BUSY_OUT),
.CHANNEL(CHANNEL_OUT[4:0]),
.DO(DO_OUT[15:0]),
.DRDY(DRDY_OUT),
.EOC(EOC_OUT),
.EOS(EOS_OUT),
.JTAGBUSY(),
.JTAGLOCKED(),
.JTAGMODIFIED(),
.OT(OT_OUT),
.MUXADDR(),
.VP(VP_IN),
.VN(VN_IN)
);

endmodule

```

Example Design Test Bench

This subsection provides a simple test bench that sets up a 50 MHz DCLK and then reads the status registers (conversion results) at the end of a sequence when EOS goes High. The test bench also monitors the alarm outputs so alarm events can be seen. Analog signals are not introduced into the simulation using the test bench. Analog signals are read from a file by the simulation model. The SIM_MONITOR_FILE attribute used in the XADC instantiation points the model to the location of this file, known as the Analog Stimulus file. The analog stimulus file used for this example is shown in Figure 6-6.

```

1 // This is an example of an Analog Stimulus File
2 // Used to inject analog signals (e.g.,volts) input the simulation
3 // Time is in nano seconds & inputs are volts or degrees C for temperature
4 // In this example the Vccaux supply moves outside the 1.89V upper alarm limit at 27us
5 // An alarm is generate when the Vccaux channel is sampled and converted by the ADc
6
7 TIME VAUXP[0] VAUXN[0] VAUXP[1] VAUXN[1] VAUXP[2] VAUXN[2] VAUXP[3] VAUXN[3] Temp VCCINT VCCAUX VBRAM
8 00000 0.005 0.0 0.001 0.0 0.5 0.0 0.1 0.0 25 1.0 1.8 1.0
9 27000 0.020 0.0 0.022 0.0 0.49 0.0 0.2 0.0 85 1.05 1.9 1.05
10 50000 0.049 0.0 0.051 0.0 0.51 0.0 0.5 0.0 50 0.95 1.71 0.95
11 73000 0.034 0.0 0.067 0.0 0.53 0.0 0.0 0.0 60 1.00 1.8 1.0

```

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Figure 6-6: Analog Stimulus File

```

`timescale 1ns / 1ps

////////////////////////////////////
// Company:  Xilinx Inc.
//
// Create Date:    09/02/2010
//
// Module Name:    My_XADC_tb
// Project Name:    XADC UG Example Test Bench

```

```
// Target Devices: Virtex-7
// Tool versions: 13.0.2 Beta 2
// Description: Test bench for XADC UG Example design
//
// Dependencies:
//
// Revision: 1.0
// Comments: See 7 Series FPGAs XADC User Guide
//
/////////////////////////////////////////////////////////////////

module My_XADC_tb;

    // Inputs
    reg VAUXP0;
    reg VAUXN0;
    reg VAUXP1;
    reg VAUXN1;
    reg VAUXP2;
    reg VAUXN2;
    reg VAUXP3;
    reg VAUXN3;

    reg [6:0] DADDR;
    reg DCLK;
    reg DEN, DWE;
    reg RESET;

    // Outputs
    wire [15:0] D0;
    wire DRDY;
    wire EOS;
    wire BUSY;
    wire [4:0] CHANNEL;
    wire VCCAUX_ALARM;
    wire VCCINT_ALARM;
    wire USER_TEMP_ALARM;
    wire VBRAM_ALARM;
    wire ALARM;

    // Testbench variables

    wire GND_BIT;
    assign GND_BIT = 0;

    reg [15:0] temp_output;
    reg [15:0] Vccint_output;
    reg [15:0] Vccaux_output;
    reg [15:0] Vccbram_output;

    reg [15:0] Aux0_output;
    reg [15:0] Aux1_output;
    reg [15:0] Aux2_output;
    reg [15:0] Aux3_output;

    // Instantiate the Unit Under Test (UUT)

    My_XADC uut (
        .VAUXP0(VAUXP0),
        .VAUXN0(VAUXN0),
        .VAUXP1(VAUXP1),
        .VAUXN1(VAUXN1),
        .VAUXP2(VAUXP2),

```

```

        .VAUXN2(VAUXN2),
        .VAUXP3(VAUXP3),
        .VAUXN3(VAUXN3),
        .DADDR_IN(DADDR),
        .DCLK_IN(DCLK),
        .DEN_IN(DEN),
        .DWE_IN(GND_BIT),
        .RESET_IN(RESET),
        .DO_OUT(DO),
        .DRDY_OUT(DRDY),
        .EOS_OUT(EOS),
        .BUSY_OUT(BUSY),
        .CHANNEL_OUT(CHANNEL),
        .VCCAUX_ALARM_OUT(VCCAUX_ALARM),
        .VCCINT_ALARM_OUT(VCCINT_ALARM),
        .USER_TEMP_ALARM_OUT(USER_TEMP_ALARM),
        .VBRAM_ALARM_OUT(VBRAM_ALARM),
        .ALARM_OUT(ALARM)
    );

// Clock Generator
`define DCLK_MHZ      50
always #(1000.0 / (2*`DCLK_MHZ)) DCLK = ~DCLK;

initial begin
    // Initialize Inputs
    DADDR = 0;
    DCLK = 0;
    DEN = 0;
    RESET = 1;

    // Wait 4 us for global reset to finish
    repeat(200) @(posedge DCLK);
    RESET = 0;

    //$finish;
end

// Read back results when EOS pulses
// indicates end of sequence.

always@(posedge EOS) begin

    drp_read(0);          // Read the result for temp
    temp_output = DO;     // Save the result for temp

    drp_read(1);          // Read the result for Vccint
    Vccint_output = DO;   // Save the result for Vccint

    drp_read(2);          // Read the result for Vccaux
    Vccaux_output = DO;   // Save the result for Vccaux

    drp_read(6);          // Read the result for Vccbram
    Vccbram_output = DO;  // Save the result for Vccbram

    drp_read(16);         // Read the result for Aux0
    Aux0_output = DO;     // Save the result for Aux0

    drp_read(17);         // Read the result for Aux1
    Aux1_output = DO;     // Save the result for Aux1

    drp_read(18);         // Read the result for Aux2
    Aux2_output = DO;     // Save the result for Aux2

```

```

drp_read(19);          // Read the result for Aux3
Aux3_output = D0;      // Save the result for Aux3

end

// Helper tasks - to read DRP Status Registers
// Note: this task assumes there are variables defined: DCLK, DADDR, DEN, DWE, DRDY

task drp_read;
input [6:0] addr;
begin
    @(negedge DCLK);
    DEN  = 1'b1;
    DWE  = 1'b0;
    DADDR = addr;
    @(negedge DCLK); // DEN only pulsed for one clock cycle
    DEN  = 0;
    wait(DRDY==1);    // Wait for the drdy output to go
                      // high before exiting.
    @(negedge DCLK);
end
endtask

endmodule

```

In this example, the voltage on V_{CCAUX} moves from nominal supply (1.8V) to 1.9V at 27 μ s from the start of the simulation (see [Figure 6-6](#)). The V_{CCAUX_ALARM} output goes active High shortly after this event when the V_{CCAUX} supply is measured during the sequence (see [Figure 6-7](#)).

Simulation Output

The simulation output shown in [Figure 6-7](#) shows two passes through the user-defined sequence in continuous sampling mode. The channels monitored in the sequence can be seen by looking at the CHANNEL[4:0] bus. The sequence is 8, 0, 1, 2, 3, 6, 10, 11, 12, and 13, which corresponds to Calibration, Temperature, V_{CCINT} , V_{CCAUX} , V_{CCBRAM} , AUX0, AUX1, AUX2, and AUX3. Then the sequence repeats. The calibration channel takes longer to complete than the other channels because the calibration routine involves three conversions (measurements) using the ADC.



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Figure 6-7: XADC Simulation Output

When the EOS signal pulses High for one DCLK period at the end of a sequence the test bench reads the Status registers. The simulation model used the full 16 ADC conversion result because it is an ideal model of the ADC. Thus, for example, the result for the first V_{CCINT} measurement ($Vccint_output[15:0]$) is 5555h, which corresponds to 1V. In [Figure 2-9, page 29](#), 1V is shown as 555h. This is a 12-bit MSB justified result. However, the four LSBs of the Status register also contain data that would be 5h if the ADC was an ideal 16-bit ADC.

VCCAUX_ALARM output goes High during the second pass through the sequence. ALARM also goes High because it is a logical OR of all the alarm signals. Other alarm events can be added by editing the analog stimulus file and rerunning the simulation.

